

GEOLOGY OF VENEZUELA AND ITS OIL FIELDS*

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INTRODUCTORY NOTE

During September, 1951, a National Petroleum Convention was held in Caracas under the auspices of the Venezuelan Government with the cooperation of the Venezuelan oil industry. The meetings were attended by a great number of prominent oil men from throughout the world. The success attained by the convention was due in major part to the officials of the Ministerio de Minas e Hidrocarburos of Venezuela.

Eight papers, dealing with various phases of the Venezuelan oil industry, were presented at the convention and have been published in English as well as Spanish by the Oficina Técnica de Hidrocarburos of the Ministerio de Minas e Hidrocarburos in a volume entitled *Text of Papers Presented at the National Petroleum Convention, Caracas, Venezuela, September 9-18, 1951*. Because of the widespread interest in Venezuelan oil, it has been thought worth-while to reprint Chapter One, "Geological Review," in this *Bulletin*.

Grateful acknowledgment is made to Santiago E. Vera Izquierdo, Minister of Mines and Hydrocarbons, and to Gustavo Thery Fombona, director, F. Medina Olivieri, Luis Jacobo Cordero, and Domingo Noriega Salazar of the Oficina Técnica de Hidrocarburos, for their kind cooperation in the republishing of this paper.

Thanks are also due to Creole Petroleum Corporation, Mene Grande Oil Company, Shell-Caribbean Petroleum Company, Socony-Vacuum Oil Company, Texas Petroleum Company, and Venezuelan Atlantic Refining Company for their contributions to this paper and their permission for its publication.

INTRODUCTION

It is by now common knowledge that the present-day configuration of the lands and seas, the mountains and valleys, the hills and plains, has not existed from the beginnings of time, but is the result of a gradual development down through the ages. It is the function of the geologist as a scientist to reconstruct the history of a region by tracing backward step-by-step the changes that have occurred—the shifts in seas and lands, the risings and lowerings of mountains, and the evolution of basins and plains. Furthermore, it is the function of the geologist as a member of the petroleum industry to discover new oil accumulations and to aid in the development of those already known. The origin, migration, and accumulation of oil are closely related to the geologic history of the area where it occurs, to the nature of the sediments in which the oil originated and in which it subsequently accumulated, and to the foldings and breakings of strata that have resulted in the formation of structures suitable for such accumulation.

The first part of this paper is an attempt to decipher the general geological history of Venezuela as a setting for the conditions under which the oil-bearing strata originated. Such an analysis may be accomplished through a study of the rock strata visible in the mountains or in samples taken from wells, of the structures in which these strata are found, and of the relationships of the strata. Knowledge of the remains of organisms (fossils) in the rocks not only tells us their relative age but also something of their environment of origin; whether marine, brackish water, continental, *et cetera*. As an example of this method of analysis, let us consider the principal oil-bearing section in the Jusepin field which consists of a series of grayish shales and silty shales with local lenses of fine sandstone called the La Pica formation. Fossils are present but not abundant. The La Pica sediments do not lie transitionally above the underlying rocks but make an angle with those underneath. The formation above the La Pica also shows an angular (unconformable) relationship in the north but this angularity disappears southward. From these data, the geologist draws the following inferences. 1. The La Pica

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formation of Jusepfn originated along the edge of a marine basin, the La Pica beds themselves being deposited close to the shore in brackish to shallow marine waters; this fact is shown both by the type of sediment and the fossils. 2. The La Pica is of early and middle Miocene age, as indicated by fossil evidence. 3. Before the existence of the La Pica sea the Jusepfn area was dry land and exposed to erosion (indicated by unconformity with the underlying strata). 4. After the deposition of the La Pica sediments, the area was again uplifted, becoming dry land and exposed to erosion in the north (unconformity with overlying strata in north), but not in the south. The oil field of Jusepfn has been dependent on these events. The oil probably originated in the marine shales and accumulated in the lensing sandstones. Later tilting permitted concentration of oil and gas in an updip position and the underlying unconformity prevented their escape into surrounding rocks.

One of the essentials to an analysis of the geologic history of any area is the correlation of formations, showing which formations originated at the same time in various parts of the area, so that comparisons can be made of the events taking place contemporaneously throughout the region. Because no up-to-date correlation chart was in existence for Venezuela, a committee was appointed to prepare one for the National Petroleum Convention. This committee consisted of: E. Mencher, Socony-Vacuum Oil Company, chairman; K. P. Dallmus, Creole Petroleum Corporation; H. J. Fichter and P. de Schumacher, Shell-Caribbean Petroleum Company; H. H. Renz, Mene Grande Oil Company; C. González de Juana and L. Ponte R., Ministerio de Fomento, Instituto Nacional de Minería y Geología. The committee is indebted to numerous other geologists for information included in the chart. Special recognition is given H. Alberding, J. W. Butler, G. W. Crickmay, A. N. Dusenbury, Jr., L. W. Henry, R. H. Robie, N. E. Weisbord, J. G. Wilson, and M. W. Zaikowsky.

Any paper on a scientific subject these days depends largely on data and conclusions, published or unpublished, of the great many scientists who have been in the past or who are at the present engaged in similar studies. Limitations of space prevent the listing of the names of all those whose work has been used. Specific references are given in the text and are mentioned in the bibliography at the end of this paper.

PART I—GENERAL GEOLOGY*

E. MENCHER^o

SUMMARY OF GEOGRAPHY AND PHYSIOGRAPHY

The United States of Venezuela is in northeastern South America. It is bordered on the west and southwest by the Republic of Colombia, on the south and southeast by Brazil, and on the east by British Guiana. On the north it is washed by waters of the Caribbean Sea. The total area is 912,050 square kilometers. Off the coast, there are a number of islands, of which the largest is Margarita.

Politically, Venezuela is divided into twenty states, two Federal Territories, and the Federal District. The total population is 4,985,716, according to the 1950 census.

Geologically, Venezuela is a varied and interesting country, partaking of some of the most important geologic provinces of South America. 1. The southeastern part of the country, comprising the whole area beyond the Orinoco River, is an extension of the large stable "corner stone" mass of the continent, known as the Guayana shield. 2. The western part is made up mostly of the Andean Ranges, a continuation of the mighty Andean chain of western South America. In Venezuela, the Andes consist of two prongs separated by the Maracaibo basin. 3. The north-central part has the Caribbean Coast Ranges which strike across the country to Trinidad and which some geologists believe to be genetically related to the Antillean island arcs. This varied geology has given rise to a number

* A few minor changes have been made in the original text.

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of physiographic provinces, which have been classified by Liddle (1946, pp. 1-3) as follows.

1. *The Guayana Highlands:* The Guayana Highlands is an elevated plateau of low relief averaging 350-500 meters above sea-level and occupying all of the country south of the Rio Orinoco, except for the Orinoco delta. The area is an extension of the Guayana shield and is underlain in most parts by the oldest rocks in Venezuela: pre-Cambrian crystalline gneisses and granites. Topography is due to the erosion and weathering of these relatively homogeneous rocks. In the southeast, near the Brazilian and British Guiana boundaries, a series of high, flat-topped mesas standing above the plateau is the result of the erosion of essentially flat hard Roraima sandstones capping igneous rock.

2. *The Venezuelan Andes of the States of Táchira, Mérida, and Trujillo:* Part of one of the great mountain systems of the world, the Venezuelan Andes attain an elevation of 5,007 meters (16,427 feet) at Pico Bolívar. The range consists of folded and faulted strata ranging from pre-Cambrian to

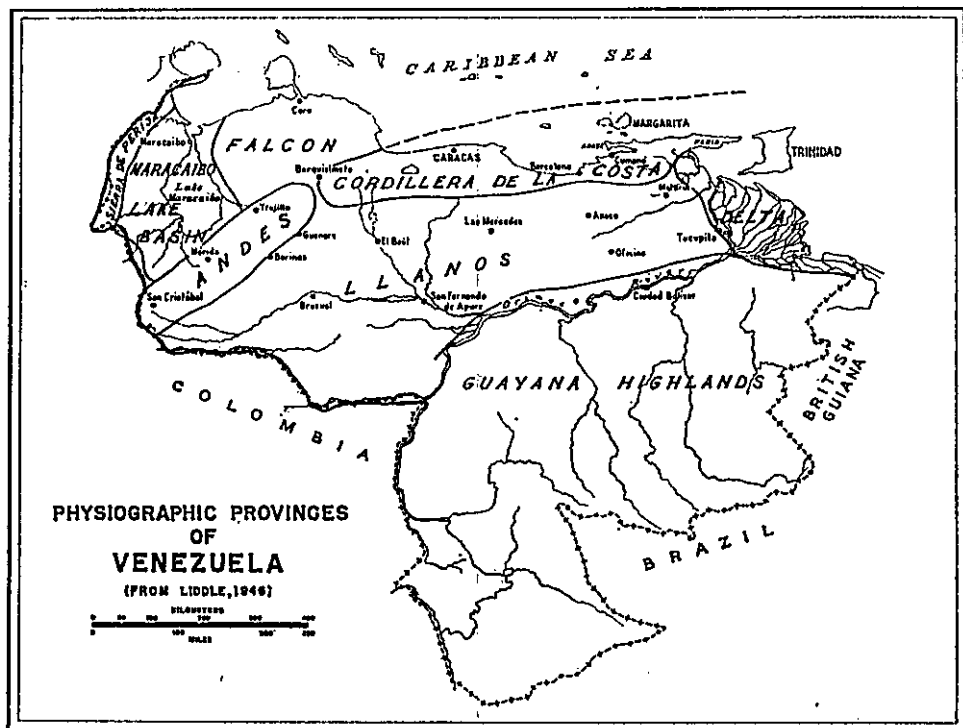


FIG. 1.—Physiographic provinces of Venezuela.

Late Tertiary in age. Elevation of the Andes occurred during the Tertiary period, reaching its culmination during the Mio-Pliocene.

3. *Sierra de Perijá:* Another branch of the Andean chain, the Sierra de Perijá is separated from the Venezuelan Andes by the Maracaibo basin. Its nature and history are similar to those of the Táchira-Mérida-Trujillo Andes. The Sierra de Perijá is at the extreme western edge of Venezuela, along the Colombian border.

4. *Cordillera de la Costa:* The Coast Range complex is an important mountain province striking in general east across the country from Barquisimeto, Lara, to the end of the Paria Peninsula and then on into the Northern Range of Trinidad. Throughout most of its length, the Cordillera borders the Caribbean Sea. Elevations are greatest (2,800 meters maximum) in the vicinity of Caracas. The range is broken east of Higuerote and disappears beneath the sea for 200 kilometers before it reappears again on the Peninsula of Araya. The reason for this break has not been ascertained; it may be due to faulting. The northern part of the range, known as the Cordillera de la Costa proper, is composed

of metamorphic rocks, a great part of which are believed to be of Cretaceous age. The Serranía del Interior, a lower range on the south, is best developed in the area east of Barcelona, where it consists of highly deformed, but unmetamorphosed Cretaceous and Tertiary strata. Farther west, in north-central Venezuela, the Serranía del Interior contains an extensive belt of slightly metamorphosed to unmetamorphosed Cretaceous and Paleocene volcanic rocks in addition to the sediments. Generally considered to be an extension of the Andean system (Liddle, 1946, p. 47), the Cordillera de la Costa has an early history quite different from that of the Andes. Middle to Late Tertiary uplifts, however, appear to be related to those of the Andes.

5. *The Llanos*: The llanos constitute the great central-plain area of Venezuela which lies between the Guayana Highlands on the southeast and the Andes or Serranía del Interior on the west and north. In general, the llanos surface lies below the 200-meter level with large regions between sea-level and 100 meters. In Guárico, Anzoátegui, and Monagas, above the llanos level are extensive flat areas bounded by sharp cliffs, or "farallones." These "mesas" may reach elevations ranging from 400 to 500 meters. The llanos are underlain by essentially undisturbed Tertiary and Quaternary soft sediments. Extensive zones of alluvial deposits are still being spread out over its surface by the flooding of the Apure-Orinoco drainage system. The mesas represent the eroded remnants of older alluvial deposits uplifted in Late Pleistocene or Recent time.

6. *Maracaibo Lake Basin*: Between the Táchira-Mérida-Trujillo Andes and the Sierra de Perijá lies a low rectangular area occupied by Lake Maracaibo in its central part. The Lake is in communication with the Caribbean Sea by means of a narrow channel into the Golfo de Venezuela. The Maracaibo basin has had a long history as a depressed area.

7. *Falcón Region*: North of the Andes and the western Cordillera de la Costa is a belt of plateaus and hills comprising almost the entire State of Falcón and the northern part of the State of Lara. In the west, where they border the Maracaibo basin, the ridges are highest, the maximum elevation being about 1,900 meters at Cerro Cerrón at the western end of the Loma de Caballo. Here the ridges strike almost due north, but this direction changes sharply northeast and finally, in eastern Falcón, almost due east. The Falcón region is underlain by Tertiary rocks which were elevated into elongate ridges by intense folding and thrusting during the Andean orogeny.

8. *Delta Region*: The delta of the Orinoco River is an extensive triangular swamp more than 400 kilometers wide and reaching inland a maximum of 200 kilometers. It is built up of Recent and possibly Pleistocene deltaic deposits, but earlier sediments are related either to the llanos province of the eastern Venezuelan basin or, in the north, to the Serranía del Interior.

9. *Islands of the Caribbean Sea*: Geologically, most of the Venezuelan islands north of the mainland belong to the Cordillera de la Costa province.

GEOLOGY

PRE-CRETACEOUS

Little is known about the history of geologic events in Venezuela prior to the Late Mesozoic period. The Guayana Highlands, comprising nearly half the total area of country, are underlain in their entirety by old rocks, but, because of the lack of petroleum prospects and the difficulties of exploration, relatively little is known of the region in detail. With the recent interest in gold, diamond, and iron deposits of the Guayana, it is to be hoped that knowledge of this area will be increased. In the Andes, too, there are large areas of old rocks, which have received but cursory examination. Since only a few disconnected facts are known about Venezuelan geology before the beginning of the Cretaceous period, and since as yet no petroleum prospects have been found in the pre-Cretaceous rocks, the whole geologic time interval to the Cretaceous is discussed in a brief manner.

Presumably the oldest rocks of Venezuela are those of the Guayana shield. Their age has not been determined in Venezuela, but comparison with similar rocks farther south in Brazil indicates that they are Archean. Contorted gneisses and gneissoid granites constitute the major part of the Guayana basement complex (Liddle, 1946, p. 59), but there are some silicified porphyritic volcanic rocks near the Brazilian border. In the northern part of the Guayana, not far

south of the Orinoco River, are the ferruginous quartzites and iron concentrations of the Imataca formation (Zuloaga and Tello, 1939, pp. 398-430), which have begun to play an important part in the economy of the country. The age of the Imataca rocks is unknown and even their contact with the gneisses is obscured by the presence of an intrusive granite (Zuloaga and Tello, 1939, p. 412). All these rocks are cut by intrusions of gabbros and basaltic rocks.

Like other shield areas of the world, the Guayana remained as a positive stable element throughout most of later geologic time. There is record of only two groups of sediments having been deposited upon the shield after its formation, and neither of these is of marine origin: the first is a group of volcanic rocks occurring near Callao which has been called the Pastora series (Newhouse and Zuloaga, 1929, pp. 797-810); the second is the Roraima formation present in the Gran Sabana region and in the surrounding parts of British Guiana and Brazil. The Roraima beds are composed of white to pale purple sandstones and quartzitic sandstones, conglomerates, and shales, and red and green jasperized tuffs intruded by gabbro and diorite. The Roraima was deposited on a vast alluvial plain at a time when volcanoes were active near-by. Unfortunately, there is at present no means of knowing when this time was, although lithologically the Roraima formation most closely resembles the early Mesozoic La Quinta formation. The Roraima beds, totaling at least 2,400 meters in thickness, still retain the essentially horizontal position in which they were laid down. The present-day diamond placer deposits of the Gran Sabana appear to have had their origin in the erosion of the basal beds of this formation.

Several wells north of the Orinoco River have penetrated granites directly below Cretaceous sediments but, in several places in southern Anzoátegui and Guárico (Figs. 8A-8B), pre-Cretaceous sediments occur. The most widely distributed of these is a thick (at least 500 meters) section of homogeneous, tough, massive, greenish gray, dense, hard siltstone containing a few thin layers of calcareous glauconitic sandstone. These siltstones constitute the Carrizal formation (Hedberg, 1940, p. 201) and while they are of marine origin, and a few fossils have been found in them, their age is unknown. Underlying the Carrizal beds in the Suata and Hato Viejo wells are pinkish gray and gray sandstones and conglomerates of the Hato Viejo formation.

Pre-Cretaceous rocks appear in an isolated group of hills at El Baúl, Cojedes. They consist of pink granite and gray syenite, light gray to red phyllites and quartzites, rhyolite, and trachyte. The phyllites and quartzites obviously had their origin in sediments that were subsequently metamorphosed. It is not certain whether the granite is younger or older than these sediments. The trachyte intrudes the phyllites and is the younger, while youngest of all is the rhyolite, which had its origin in lava flows.

The central part of the Andes is made up of extensive gneisses, schists, and other types of crystalline metamorphic rocks, and granites (Kündig, 1938, p. 22), which have been elevated by the Andean orogeny. All are probably of pre-

Cambrian age. Younger than the crystalline metamorphics is a series of gray schistose shales and slaty shales and dark gray and black phyllites and slates. South and southwest of Mérida, these rocks have been called the Mucuchachí series (Christ, 1927, p. 397; Kehrer, 1938, pp. 54-55); at the northeast, similar rocks are placed in the Rio Momboy series (Liddle, 1946, pp. 114-18). Fossils have been found in the Rio Momboy rocks which date them as Devonian. Older fossils (Ordovician) have been collected from soft white silty shale (Leith, 1938, pp. 337-44) associated with more metamorphosed rocks, but the relationship of the Ordovician rocks to the Mucuchachí phyllites is obscure. A good section of fossiliferous Devonian strata, the Rio Cachirí series, occurs also in the Sierra de Perijá (Liddle, 1946, pp. 109-14, 118-20). Here the beds are less metamorphosed and consist of gray sandy ferruginous shale and micaceous ferruginous sandstone, black shale, blue and black limestones, gray quartzite, and red shale.

Fossiliferous sediments of Carboniferous and Permian age have been observed in a number of limited localities in the Andes (Kehrer, 1938, pp. 52-54). They have been named the Palmarito series and are composed of soft, blue-gray shale and massive, dark gray to blue-black, hard limestones (Sutton, 1946, p. 1636). The lower part of the section contains gray slates and argillites with subordinate amounts of green and red slate. The Palmarito is also present in the Sierra de Perijá (Liddle, 1946, pp. 127-30; Sutton, 1946, p. 1636) where it consists of red micaceous shale and sandy shale with some interbeds of crinoidal limestone. The Permo-Carboniferous probably lies with unconformity on the Devonian. Dikes and sills of rhyolite, andesite, and basalt intrude the Palmarito sediments.

Despite the paucity of our information on Paleozoic sedimentation in Venezuela, certain tentative conclusions can be drawn concerning the history of that era. There is no question that a sea covered western Venezuela, from at least the Andes to the Sierra de Perijá, during much of Paleozoic time. That this sea may have reached central Venezuela is attested by the phyllites at El Baúl and low-grade metamorphosed sediments encountered in some of the wildcat wells northeast of Calabozo. Lithologically, the only rocks in Venezuela comparable with the phyllitic series of El Baúl are those of the Palmarito. Moreover, the relationship of the El Baúl phyllites to the trachytes and rhyolites is like that of the Palmarito to its associated igneous rocks. The Paleozoic sediments may well have been deposited in a geosyncline where disturbance and sedimentation were taking place simultaneously. (For the history of a similar geosyncline, see the later discussion of the Coast Range geosyncline during Late Cretaceous.) The geosyncline may have extended southward into Colombia. The main deformation was probably near the end of Paleozoic time, after the deposition of the Palmarito. The center of deformation lay well east of the Andes, as is shown by the fact that the Andean Paleozoic rocks lie at the very edge of the zone of metamorphism while the Sierra de Perijá Paleozoics, still farther west, are unmetamorphosed.

Following the deformation of the Paleozoic rocks, their uplift into mountains, and their partial erosion, new inland basins of sedimentation were established in

western Venezuela in the Early Mesozoic. These basins were filled with alluvial sediments, most of which have a red color and which are derived from the erosion of the Paleozoic mountains. La Quinta formation is the name which has been given to this redbed group. During and perhaps shortly after the deposition of La Quinta, magmas of basaltic character intruded the red sediments in the form of sills and dikes. Some of the igneous material penetrated to the surface, where it welled out in the form of basaltic lava.

After the deposition of the La Quinta red sediments and volcanic rocks, new uplift and disturbance of western Venezuela took place, although not on a scale comparable with that at the end of the Paleozoic era. The La Quinta beds were folded but they were not metamorphosed. Erosion soon thereafter began to level the hills and fill the valleys until, prior to the beginning of the Cretaceous period, practically the whole of Venezuela was cut down to a low plain, or peneplain.

EARLY CRETACEOUS

With the Cretaceous system we are on firmer ground in our knowledge of the geology of Venezuela. Cretaceous deposits occur throughout great parts of the country and their distribution indicates that, during Cretaceous time, at least half of present-day Venezuela was covered by a shallow sea.

About the beginning of Cretaceous time, Venezuela must be pictured as a low region of small relief. There were no Andes and no Coast Ranges. Soon after the beginning of the Cretaceous, a shallow sea began to invade the country, spreading southward from the Caribbean. This transgression may have started even before the Cretaceous in the area north of the present Venezuelan coast, for ammonites have been found in the Northern Range of Trinidad which suggest very late Jurassic age for some of the metamorphosed limestone there. On the mainland, however, no strata older than Cretaceous have been identified. The encroachment of the sea upon the land was very gradual, with on-shore and close-to-shore sediments and swamp deposits slowly giving way to more marine beds as time progressed. The early Cretaceous deposits in the northeast, consisting principally of cross-bedded, coarse sandstones, and sandy carbonaceous shales below and of reefal limestones and marine shales above, have been called the Barranquín, El Cantil, and Chimana formations; similar deposits in the Andes and the west constitute the Cogollo group.

By the end of Early Cretaceous time the shoreline seems to have advanced to a little less than midway between the present coast and the Orinoco River. Westward, it probably passed just south of the Mercedes oil field and north of the El Baúl region, although data are extremely meager. From El Baúl, the shoreline curved southwest, and was located somewhat east of Bruzual and Guasdalito. In general, then, the shoreline paralleled the present outlines of the Guayana Shield, but the shield was more extensive.

The sea continued its invasion during much of Late Cretaceous time. Although there is little evidence for a depositional break between the Lower Cre-

taceous formations and the Upper Cretaceous, the contact between them is remarkably distinct. This, taken with the widespread and uniform character of the dark marine shales and limestones of the earlier beds of the Late Cretaceous deposition, suggests that, near the end of the Early Cretaceous, there may have been a gentle but very widespread deepening of the sea, so that open-sea conditions prevailed over vast areas where formerly only close-to-shore and brackish

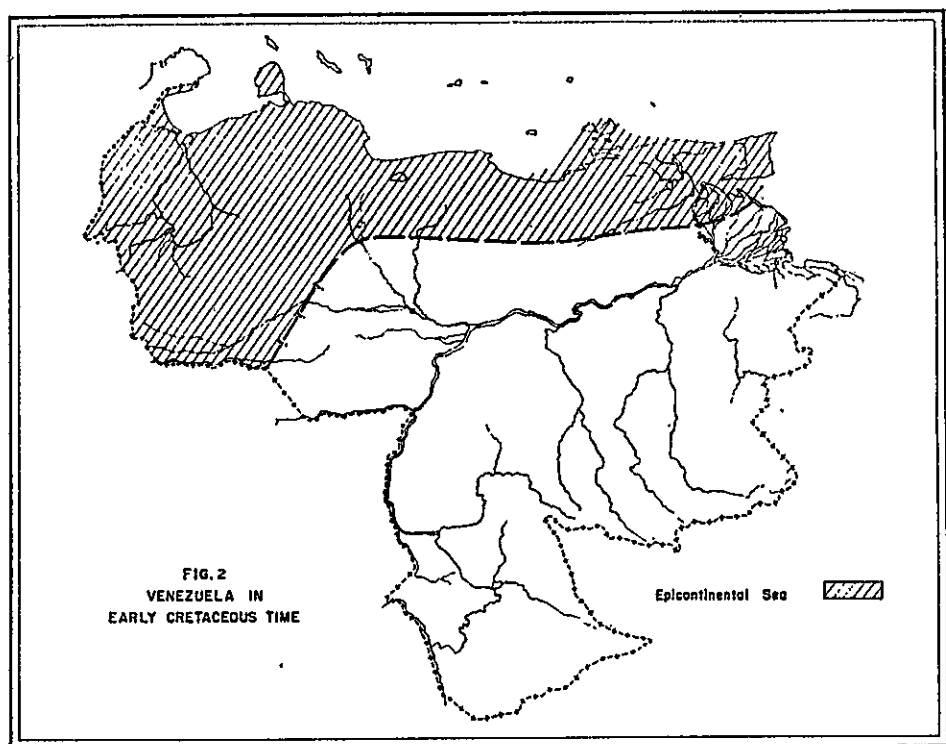


FIG. 2.—Venezuela in early Cretaceous time.

sediments were laid down. How far southward the Late Cretaceous shoreline advanced is difficult to estimate. Marine sediments in wells of the Oficina area show that this area was inundated during part of Late Cretaceous time, but Cretaceous strata are either completely missing or do not show marine character in wildcat wells south of the general Oficina area. It is therefore probable that the greatest advance of the Cretaceous sea did not reach the present course of the Orinoco River.

The extreme homogeneity of deposition over wide areas, the thin and uniform bedding, the predominance of dark, dense, fine-grained limestones and dark platy shales, all rich in the remains of pelagic organisms intimate, however, that these

open-sea conditions were no ordinary ones. Rather, it would appear that deposition was exceedingly slow and stable during long periods of time, that the land at the south was low, and that deposition took place under somewhat reducing conditions. These reducing conditions may have been caused by depth or by abundance of organic matter being brought into the sea from extensive swamps along its margin. In northeastern Venezuela, these limestones and shales are about 740 meters thick and are called the Querecual formation. The corresponding sedi-

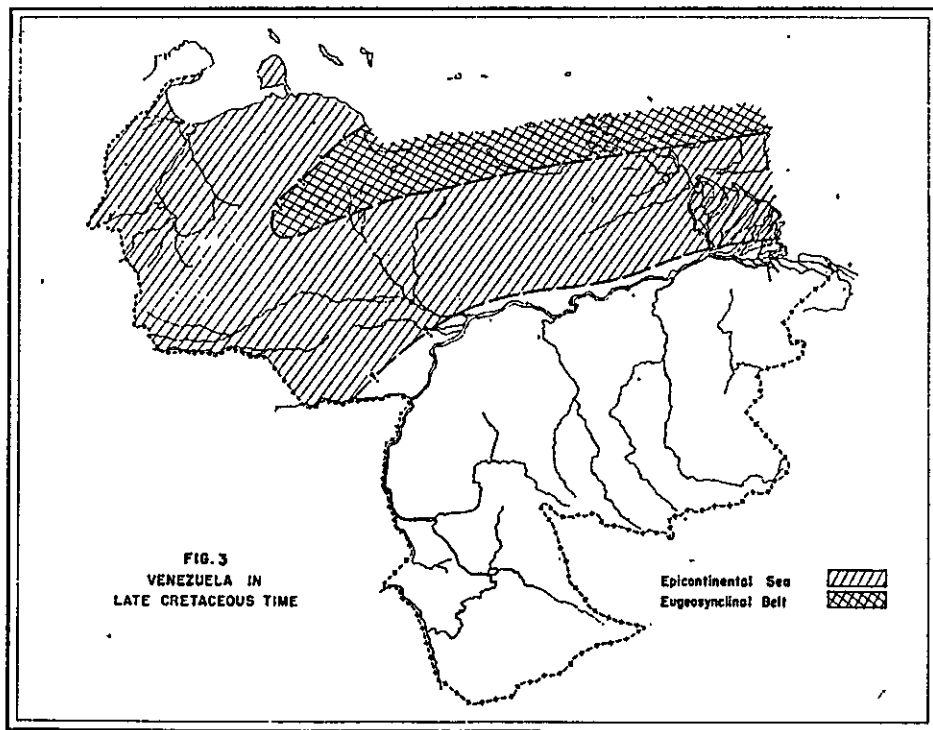


FIG. 3.—Venezuela in late Cretaceous time:

ments in western Venezuela are considerably thinner, with a maximum of about 315 meters in the Carache Valley of Trujillo, and have been assigned to the La Luna formation.

LATE CRETACEOUS TO PALEOCENE

The Cretaceous formations so far described consist of sediments of epicontinental and miogeosynclinal origin. Such deposition continued to the end of the Cretaceous in the Andes as well as in south-central and eastern Venezuela, although there was a constant shallowing of the area from late La Luna time onward.

In north-central Venezuela, on the other hand, conditions began to change.

After deposition of black shales and limestones corresponding with the lower part of the La Luna formation, the area east of Barquisimeto and extending through Margarita, Araya, and Paria to Trinidad and Tobago became a zone of great instability with large and recurrent diastrophic movements accompanied by igneous activity. This zone represents a true geosyncline ("eugeosyncline" of Stille and Kay or "tectogene" of Hess). That the eugeosyncline's western limit did not reach Lake Maracaibo or the Andes proper (Serranía de Trujillo) is shown by the nature and thickness of the Late Cretaceous deposits (La Luna-Colon-Mito Juan) there. The southern boundary extended roughly from Guanare northeastward through Ortiz to about Piritú and from the Golfo de Cariaco across northern Trinidad and into the Atlantic Ocean. At the north, the area now uplifted into the Cordillera de la Costa of central Venezuela, the peninsulas of Araya and Paria, and the islands of Margarita and Tobago was all included within the geosynclinal limits. The relationship of the Dutch islands, Curacao, Aruba, and Bonaire, and the Venezuelan islands of Las Aves, Los Roques, Orchila, and Blanquilla to the eugeosyncline proper is difficult to ascertain.

The precise time of origin of the eugeosyncline is still open to question. The event may have taken place about the beginning of Coniacian time for the following reasons. 1. At the foot of Cerro Garrapata, a few kilometers northeast of San Juan de los Morros, is a series of black silty shales and graywacke sandstone interbedded with submarine lava flows, volcanic conglomerates, and agglomerates. The finding of the ammonite *Peroniceras* in the shale dates these deposits as Early Coniacian. Structural complications prevent certainty of the position of these beds within the general succession of the area, but it is probable that they are near the base of the volcanic-sedimentary series which developed within the eugeosyncline. 2. The widespread distribution of Late Cretaceous bedded cherts throughout Venezuela, in the upper La Luna formation of western Venezuela and in the San Antonio formation in the east. Evidence in many places in the world indicates a genetic relationship between volcanism and bedded cherts; the writer thinks this relationship holds also in Venezuela. Nevertheless, only locally, as in various places in the vicinity of the road between San Sebastián and Valle de Morín in southern Aragua and on Margarita, are cherts found interbedded with volcanic rocks. Nor are those the areas of thickest chert sections. Rather, the best development of the cherts seems to have been in the miogeosynclinal areas adjacent to the eugeosyncline. Here colloidal silica, originating in the submarine volcanoes of the eugeosynclinal zone, was slowly deposited along with pelagic limestones of the La Luna-Querecual type and black carbonaceous shales. In the eugeosyncline, itself, conditions were too unstable for the formation of appreciable thicknesses of pure chert. Instead, the silica mixed in with clastic deposits, resulting in the hard, partly siliceous shales and quartzites of the Arayanes group of north-central Venezuela.

A geosyncline, such as the one postulated across northern Venezuela for Late Cretaceous time, is an extremely unstable zone. Not only is there sinking and

deepening of the belt as a whole but, within its limits, there are compression and buckling, so that deformation proceeds concomitantly with sedimentation. Meanwhile volcanoes add their materials in the form of flows, tuffs, and volcanic sediments. Because of the buckling, areas lately beneath the sea are uplifted into islands or geanticlinal ridges and recently deposited strata are deformed, then eroded and redeposited, all within the confines of the geosyncline. Thus, sedimentation is variable and conglomerates with pebbles and boulders of previous geosynclinal sediments are common. During the history of such a geosyncline, a time may come when the compressional forces are strong enough to cause metamorphism of a part of the then-existing sediments and igneous rocks and at this same time, or later, the metamorphosed rocks may be thrust platformward so that rocks from the inner parts of the geosyncline may lie above or in contact with rocks of the same or younger age from the outer parts or even with rocks of the platform. The difficulties of unraveling the history of such an area can well be appreciated.

Before Coniacian time, then, northern and western Venezuela may be visualized as covered by a sea almost to the present location of the Orinoco River. From Coniacian time onward throughout the rest of the Cretaceous, the sea slowly retreated northward and westward but, at the same time, there developed a narrow unstable belt across northern Venezuela from Barquisimeto through Araya and Paria to Trinidad and out into the ocean. While most of this zone remained beneath the sea, parts of it were periodically uplifted into islands. Volcanoes built themselves up from the sea floor until some reached the surface, forming volcanic islands. Earthquakes were common. This north-central area of Venezuela partook of many of the characteristics associated with present-day earthquake and island arc zones, such as that in the Pacific Ocean off the coast of Asia.

In the stable areas, the Upper Cretaceous sediments consist in large part of shales, but limestones and sandstones are not uncommon. As already pointed out, cherts and silicified beds commonly occur, principally in the lower part of the section (San Antonio and La Luna formations). In the west, the La Luna formation is succeeded by 350-750 meters of shales of which the lower part, known as the Colón formation, consists almost entirely of massive dark gray to black hard pyritic shale. There is no obvious break between the La Luna and Colón formations but in many places a thin glauconitic sandstone containing water-worn shells separates the two formations. It is thought that this bed may indicate a gap in deposition between the two formations. In the area west of Lake Maracaibo, there is a 20-meter thickness of gray limestone just above the glauconitic sandstone, locally known as the Basal Colón, or Socuy, limestone, which is of importance as an oil producer.

The change from Colón to the overlying Mito Juan shales is very gradual, and in many places it is impossible to draw the line between the two. The Mito Juan is characteristically sandier than the Colón and is composed chiefly of silty to sandy shales, commonly greenish, interbedded in their upper part with sand-

stones and impure limestones. The upper part of the Mito Juan is indicative of brackish-water conditions and shows that the Cretaceous sea was withdrawing from western Venezuela. However, this withdrawal does not seem to have been accompanied by any folding of the rocks. Rather, some tilting probably occurred, with movement of blocks along faults.

In eastern Venezuela, the Querecual limestones and shales are succeeded without apparent break by somewhat similar limestones and shales, but interbedded with these are abundant cherts and some sandstones which constitute the San Antonio formation. The possible relationship of the cherts to volcanism in north-central Venezuela has already been mentioned. Sandstone beds become more numerous and thicker upward, until, in general, a solid 30-300-meter thickness of fine-grained, gray, quartzitic sandstone is attained. This is the San Juan sandstone and marks the base of the Santa Anita group. In some areas, such as near Barcelona, the San Juan sandstone is not present, but is replaced by shales with some thin sandstones. Deposition was continuous from the San Antonio into the Santa Anita group.

Above the San Juan sandstone, where present, or directly above the San Antonio formation, where the sandstone is absent, is a section of Vidoño dark gray foraminiferal glauconitic shales with a few minor beds of sandstone and hard calcareous siltstone. These shales have a maximum thickness of about 500 meters near Barcelona. They are comparable with the Colón-Mito Juan series in the west but do not necessarily indicate a general shallowing of the sea.

Thus, sedimentation on the epicontinental areas of both eastern and western Venezuela shows no abrupt changes to the end of the Cretaceous. In the west, however, a gradual retreat of the sea began about the same time as the eugeosyncline was initiated in north-central Venezuela, while in the east the incursions of sands beginning in the San Antonio formation and reaching a peak in the San Juan sandstone suggest that movements within the geosynclinal area may have had their effect on epicontinental sedimentation.

Sedimentation in the eugeosyncline consisted of the dumping of masses of sediments mixed with lava flows and igneous materials in the unstable zone. The source of most of these sediments was in the newly risen islands within the zone itself, from the volcanoes which developed on the sea floor, and possibly from land areas on the north, which are now beneath the sea. The deposits consist, therefore, of conglomerates, coarse- to fine-grained graywacke sandstones and quartzites, and silty shales, all generally dark gray to black. Interbedded with these are flows of basaltic to andesitic composition, tuffs, and agglomerates. It is suggested the name "Arrayanes" be applied to this sedimentary-volcanic series. The total thickness of the Arrayanes is not known; it has been so broken by folding and faulting that more study is required before the details of deposition can be determined.

The great compressional forces that were active in the geosyncline caused the crumpling and metamorphism of a considerable part of the sediments and

volcanics of the area and their resultant conversion to metamorphic rocks: schists, quartzites, quartzite conglomerates, marbles, "greenschists," *et cetera*. These metamorphics now comprise the principal rocks of the Cordillera de la Costa and of the Serranía del Interior near Caracas, the former consisting mostly of metamorphosed sediments and the latter of "greenschists." The grade of metamorphism varies considerably; nor is there any need to suppose that the metamorphism all took place at one time; rather, it probably took place at intervals throughout the history of the geosyncline.

The eugeosyncline lasted to the end of the Cretaceous. Volcanism and igneous activity continued well into the Paleocene, but, on the whole, the Paleocene rocks, except where they occur near thrust zones, are less deformed than those in the Cretaceous. Hence, it is probable that the forces related to the geosyncline were rapidly diminishing toward the end of the Cretaceous. The Paleocene deposits (Guarico formation) of north-central Venezuela consist of reefal limestones (Morro limestone), soft olive shales, sandstones rich in fragments of igneous rocks, tuffs, agglomerates, and flows. In general, they lie unconformably on the previous geosynclinal deposits, and though they may be intensely folded locally, few are truly metamorphosed, even in regions where they overlie partly metamorphosed earlier rocks.

The Paleocene deposits of north-central Venezuela are much more restricted in their distribution than are the Cretaceous rocks, being fairly well limited to a narrow belt extending from Tinaco to Unare. The nature of the deposits and their fossils imply shallow seas, while the association of many of the large reefs with igneous rocks suggests that the sediments may have formed on the flanks of volcanoes.

With the deposition of the Paleocene, the history of the north-central Venezuelan mainland as a region of geosynclinal activity seems to have terminated completely.

In the epicontinental areas of western and eastern Venezuela, sedimentation continued from the Cretaceous into the Paleocene with but little change, although regression of the Cretaceous sea resulted in the later deposits in a shallow or brackish-water to swamp or non-marine environment. The Paleocene Guasare formation from the Sierra de Perijá to Falcón is of shallow-water marine origin and consists principally of glauconitic and sandy limestone with abundant oyster and *Venericardia* remains, interstratified with subordinate amounts of sandstone and calcareous shale (Sutton, 1946, p. 1657). Farther south, in Táchira and Mérida, the equivalent beds (lower Angostura formation) were deposited in a fresh- to brackish-water environment and contain many beds of coal, indicating that much of the area formerly covered by the Cretaceous sea had emerged.

The Cretaceous sea had retreated from central Venezuela, too, and the only Paleocene sediments are those already mentioned.

Shallow marine Paleocene is present also in northeastern Venezuela, where it is known as the Caratas formation of the Santa Anita group. It is completely transitional into the underlying Vidoño shales.

CRETACEOUS-TERTIARY BOUNDARY

It is evident that, with the exception of the eugeosynclinal belt of north-central and northeastern Venezuela, the Late Cretaceous-Paleocene interval was without special incident. True, in many places the Cretaceous seas had retreated completely from areas they formerly occupied, but this process was gradual and had commenced long before the end of the Cretaceous. It is probable that the retreat of the sea was accompanied by faulting and uplift of the platform areas, but these changes were not "revolutionary." Even in the eugeosyncline, though deformation and thrusting may have culminated at the end of the Cretaceous, volcanism and sedimentation continued into the Paleocene, albeit on a more restricted scale.

There is no evidence for a rise of the Andean region at any time during the Cretaceous or Paleocene. All the facts point to this area as being low in Late Cretaceous-Paleocene time, with its surface covered by a shallow sea, by swamps, or by alluvial sediments. However, by the end of the Cretaceous, it is probable that the Cordillera de la Costa was already in existence and, from that time onward, formed a periodic barrier between sedimentation on the mainland at the south and that in the sea-island area at the north.

EARLY TERTIARY

EARLY TO MIDDLE EOCENE

With the early Eocene, sedimentation ceased for a long time in the major part of Venezuela. On the mainland, only in the west, in the Lake Maracaibo region and along the western flank of the Andes into Falcón, have deposits of early to middle Eocene age been found. Lower and middle Eocene strata are present also on the Island of Margarita and on Trinidad.

In the western, southern, and southeastern parts of the Maracaibo Lake area, lower and middle Eocene formations indicate that they were deposited in a brackish marine to fresh-water environment. These sediments are known by a variety of formational names: Marcelina, Paso Diablo, Barco and Los Cuervos, Angostura, Ranchería, and, in older terminology, "Third Coal." All are similar, consisting essentially of interbedded light gray sandstone, in places calcareous or quartzitic, and dark gray, carbonaceous, micaceous shale with interbeddings of sub-bituminous to bituminous coal. Maximum thicknesses are about 1,000 meters. With the possible exception of the Angostura, all are gradational downward into the underlying sediments. They demonstrate again that the emergence of the land from the Cretaceous sea was gradual and of long duration. The land areas surrounding the zone of sedimentation were low and perhaps somewhat like the present-day llanos.

Unusual thicknesses of lower and middle Eocene (Trujillo) sediments were deposited in the northern Lake area where the present La Concepción, Bolívar Coastal, and Mene Grande oil fields are located. The total amount of sedimentation is not known with certainty, but it is more than 3,000 meters and perhaps as

much as 6,000 meters. Lithologically the Trujillo is not essentially different from the other lower-middle Eocene formations just described; it consists principally of a "monotonous, thick series of dark gray to black, carbonaceous and micaceous sandy shales interbedded with micaceous sandstones" (Caribbean Petroleum Company, 1948, p. 525, 588). Thin seams of sub-bituminous coal occur in the lower part at the type section; there the basal strata are indurated and cut by white quartz veins (Sutton, 1946, p. 1660). On the whole the Trujillo probably

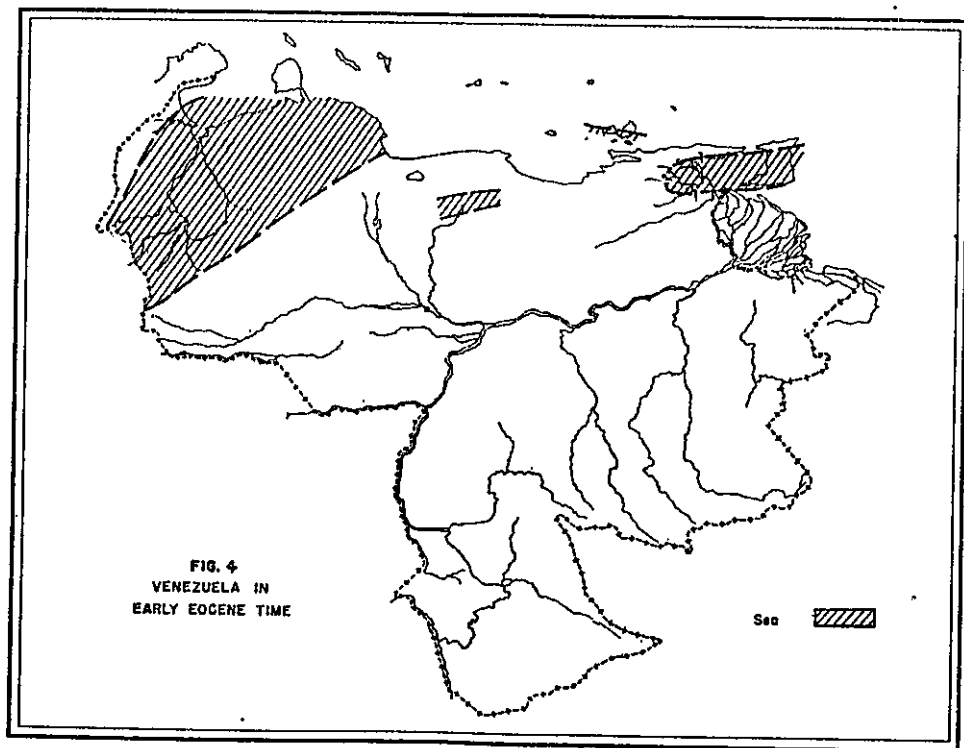


FIG. 4.—Venezuela in early Eocene time.

represents a somewhat deeper marine facies than the other formations of equivalent age at the south and west.

The great thickness of Trujillo sediments suggests deposition in a geosyncline. On the other hand, the lithologic character of the strata, the lack of igneous materials, the general conformity of the beds with the overlying Missoa formation, and the apparent absence of compressional effects indicate that the zone of deposition was not of eugeosynclinal type. It is possible that, as the Trujillo is thought to continue eastward across Falcón, where very little is known about the Eocene formations because of a thick cover of post-Eocene strata, this thick

lower-middle Eocene deposition may have been laid down in a sinking trough that connected with Margarita where there is a good marine section of comparable age.

Coarse conglomerates, lenticular algal and orbitoidal limestones, gray sandstones, interbedded calcareous gray and olive shales, argillaceous limestones, and calcareous sandstones comprise the lower to middle Eocene of Margarita. The conglomerates contain abundant boulders of unmetamorphosed volcanic and other igneous rocks, but none of this material is primary, volcanism having ceased by the early Eocene. With the exception of the uppermost shales, laid down under more open-sea conditions, all the sediments are of shallow marine origin and probably were derived from the north (K. F. Dallmuis and H. H. Renz, private communication). By assuming derivation from the north and elevation of the Coast Ranges on the south, it can be seen that the trough of deposition must have been very narrow. As mentioned previously the trough may have connected westward across the Caribbean Sea and Falcón to Lake Maracaibo.

Marine lower to middle Eocene strata occur in central and southern Trinidad. The majority belong to the Navet formation of light gray and greenish gray marls and marly clays (Cushman and Renz, 1948, p. 1). As in western Venezuela the beds grade downward without break into the Paleocene and Cretaceous. It is thought that the Navet sediments were derived from the north, from the already elevated Northern Range, which is an extension of the Coast Ranges of the mainland (Senn, 1940, p. 1564). If this is true, the lower-middle Eocene deposits of Trinidad did not connect with those of Margarita but were laid down in a separate basin which may have extended somewhat westward across the Gulf of Paria to Monagas. However, as yet no lower or middle Eocene rocks have been proved in Monagas.

By early-middle Eocene time, central and eastern Venezuela had already assumed something of the outlines of their present geographic aspects. The Coast Range was in existence across the northern edge of the continent, while the area on the south was land. Southern and central Trinidad were under water, but this embayment may have reached no farther inland than the present-day Gulf of Paria. North of the Coast Range, a trough-like sea covered part of Margarita and extended westward to Lake Maracaibo.

On the other hand, western Venezuela had a configuration far different from its present one. The Andes still had not risen and the low central Venezuelan land area sloped gradually westward and northwestward into swamps and shallow water. A deeper marine trough, however, was present in the northern Lake Maracaibo area and extended through Falcón bordering the recently elevated Coast Ranges. The whole of Falcón probably lay beneath the sea.

MIDDLE TO LATE EOCENE

The early to middle Eocene sandstone-shale formations of western Venezuela are succeeded upward, without apparent break, by a very widespread and char-

acteristic sandstone section known as the Misoa formation. The Misoa is composed "mainly of hard, quartzitic, light gray to brownish gray, medium- to coarse-grained, locally conglomeratic, thick-bedded to massive, sandstone. In the lower part of the formation are subordinate amounts of fine to medium, gray and dark gray, carbonaceous, micaceous, laminated and cross-laminated sandstone. Flagstone structure and ripple marking are common. Bedding planes are commonly defined by microscopically thin carbonaceous laminae. Interbedded with the sandstones are gray to dark gray shales that probably comprise about 15 per cent of the formation. They are generally thin-bedded, sand-laminated, highly carbonaceous, and commonly micaceous" (Sutton, 1946, pp. 1666-67). The greatest thickness recorded is 1,725 meters at Cerro Misoa east of Lake Maracaibo, in the vicinity of the Mene Grande field.

The lower part of the Misoa formation may represent the last phases of the regression that began in Cretaceous time, while the upper part is the beginning of a new transgression which reached its culmination in the late Eocene with the deposition of the marine Paují shales. Thus, in the Andean and Maracaibo Lake areas where lower to middle Eocene sediments are present, they grade transitionally upward into the sandstones of the Misoa regressive phase and these are overlain by sandstones of the transgressive phase. As a result of the new transgression, many areas that had, during Late Cretaceous or early Eocene time, emerged from the sea and undergone partial erosion, were again subjected to sedimentation. An example of such a region is the southeast flank of the Andes, where the Misoa transgressive sands lie directly on Cretaceous or early Eocene sediments. The angular difference between the Misoa and the underlying beds is, however, so slight as to be imperceptible in most places. Such apparently conformable relationship over wide areas, even where a considerable time gap is present, is again evidence that no tectonic movements of importance occurred in western Venezuela from early Cretaceous through Misoa time.

The exact age of the Misoa formation is somewhat in doubt, some geologists favoring middle Eocene and others late Eocene. It is possible that the regressive phase began in middle Eocene, while the transgressive phase did not end until late Eocene.

The Misoa formation of the eastern Lake Maracaibo shore was deposited in brackish water, while somewhat farther east, thin marine intercalations (Sutton, 1946, p. 1668) indicate a more marine environment. It is to be assumed, therefore, that marine conditions persisted in Falcón throughout Misoa time, although actually very little is known about strata of this age in Falcón.

The late¹⁰ Eocene basin in the west deepened and expanded still farther after Misoa time. The characteristic deposit of this expansion is the Paují shale. In its lower part, the Paují consists of interbedded black shales and light gray sand-

¹⁰ Late Eocene is here used in a very general sense. While majority opinion is in favor of late Eocene age for the Paují formation, some geologists, notably those of the Shell-Caribbean Company, contend that the Paují is middle Eocene.

stones. In some places, reefal limestones (Rio Caus) and glauconitic calcareous sandstones are common near the base. The main part of the Paují is made up almost entirely of dark olive-brown to black fossiliferous shales which had their origin in open-sea conditions at moderate depth.

The Paují facies is widespread in western Venezuela. Its western limit would seem to lie along an approximate north-south line that passes through the Bolívar Coastal field; at Mene Grande the Paují beds are typical and may amount to as much as 1,800 meters, while, in the Bolívar Coastal fields, only in the southern part of the Bachaquero field (the southernmost of the Bolívar Coastal fields) is the true Paují shaly facies present; the other parts of the Bolívar Coastal fields have a sandy-shaly facies, the "Upper Micaceous Sandstone" of the Shell Company (Caribbean Petroleum Company, 1948, p. 537). From Bachaquero this line passes southward to the vicinity of the headwaters of the Rio Zapa (Mackenzie, 1937, p. 280), whence it swings eastward under the llanos. North of the Bolívar Coastal fields, the Paují facies is found throughout Falcón where it is generally known as the Cerro Misión shales.

The Paují represents the most marine type of deposition during late Eocene. It grades laterally into interbedded shales, sandstones, and limestones of shallower-water origin, and these, in turn, interfinger with coal and sandy sediments of coastal swamp and fresh-water environment. The late Eocene basin, at its maximum development, may have extended as far as the Colombian border on the west and southwest and beyond the Apure River in the south. There are not enough data to fix an eastern boundary, but it is likely that it swung north and then east to connect with the late Eocene sea of northeastern Venezuela. The northern shoreline was beyond the present continental limits.

It is probable that the history of eastern Venezuela was not markedly different from that of the west during this period. As the sea had long before retreated from the mainland, no record of the middle Eocene regressive cycle is to be expected. In southern Trinidad, in the deeper part of the Trinidadian basin, the lower-middle Eocene Navet marls and clays are succeeded by the upper Eocene Hospital Hill formation which was laid down under similar open-sea conditions. Northward, however, an unconformity is developed between the Navet and overlying upper Eocene formations, showing that the middle Eocene sea must have retreated toward the end of that period, leaving some of the Navet beds exposed to erosion.

The transgressive phase in eastern Venezuela is marked by reef limestones and sandstones of the lower part of the Merecure group. These strata can be traced along the southern flank of the foothills of northern Monagas and Anzoátegui. Similar deposits (Tememure formation) occur in the northern Guarico foothills in the vicinity of Altigracia de Orituco, while shales and limestones with late Eocene faunas have been found at several places between San Sebastián and Camatagua. These sediments all had their origin along, or very close to, the shore. They are believed to have been derived from the south. Hence, their present out-

crops mark, in a general way, the southern shoreline of the late Eocene sea. Thus, it may be assumed that the Cordillera de la Costa, folded in the Late Cretaceous and Paleocene, was again partly under the sea by the late Eocene, although there are no actual outcrops of Eocene strata in the Cordilleran region.

It may further be assumed that the late Eocene sea of northeastern Venezuela connected westward with Falcón and the Paují basin of western Venezuela to give a continuous seaway from Trinidad to the Colombian border.

With the exception of the former geosynclinal areas folded during the Coast Range orogeny, the Merecure transgressive sediments lie with apparent conformity on the underlying Paleocene or Cretaceous strata, just as do the Misoa sandstones in the west. This shows again that, except for the unstable northern area, the Cretaceous regression and Eocene transgression were not separated by tectonic movements of a revolutionary character. As in the west, however, some faulting and warping took place between the two phases.

LATE EOCENE—OLIGOCENE

It is not until after the deposition of the Paují formation that evidence appears of the first rapid uplifts to affect the whole of Venezuela. At the beginning, these movements did not seriously affect the present Andean region proper, but were seemingly limited to the north and, more especially, to Falcón and the Cordillera de la Costa. The most important indication of this tectonism is unconformity between the Paují shales and the basal member (Santa Rita conglomerate) of the overlying Aguanegra formation of late Eocene age (Senn, 1935, pp. 59-62) in west-central Falcón near the Falcón-Lara boundary. Farther east, however, it is doubtful whether any true unconformity is present although there is a change upward from Paují shale lithology to shallow-water limestones (Cerro Campana limestone). On the west, it is claimed that an unconformity comparable with that at the base of the Santa Rita conglomerate occurs in the Bolívar Coastal fields (Sutton, 1946, p. 1684).

In the Andean region itself, sedimentation seems to have continued without interruption throughout most of the late Eocene. However, a pronounced shallowing of the Paují sea is shown by the incoming of ripple-marked and cross-laminated carbonaceous sandstones, oyster reefs, and orbitoidal limestones of such formations as the Medero of the southeastern foothill area and the Mene Grande of the northwestern foothills. Finally, still in late Eocene, the Medero and Mene Grande give way to the mottled mudstone, red sandstone, and conglomeratic continental sedimentation of the Parángula and Betijoque groups. In many localities this change from marine or brackish-water to continental deposition is gradual and transitional, showing that no hiatus exists between the two facies. There are some places, however, where the change is more abrupt and, very locally, a conglomerate may occur on the contact between the two facies.

It would thus appear that the first late Eocene movements were felt mainly

in the area surrounding the western extension of the Cordillera de la Costa. It is possible that these early movements were due to renewed uplift of that range.

The late Eocene disturbances just described were the forerunners of tectonic movements of greater importance. Again, these movements, now involving the folding and breaking of the previously deposited Eocene sediments, are most in evidence in the northwest, in the northern Lake Maracaibo region, western Falcón, and the Sierra de Perijá. In these areas, there is marked unconformity

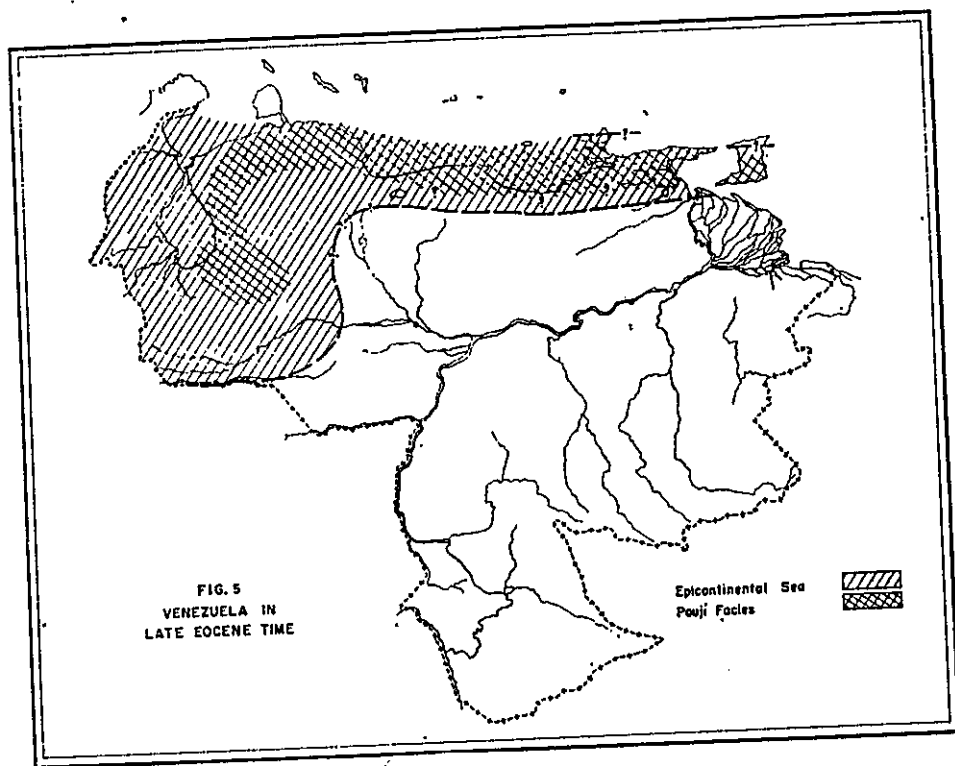


FIG. 5. Venezuela in late Eocene time.

between the uppermost Eocene strata and the overlying beds. In the Bolívar Coastal fields, and on the west side of the Lake, the intensely folded upper Eocene is overlain by non-marine *Isotea* beds which fill in synclinal areas and depressions in the Eocene land surface (Sutton, 1946, p. 1689). The age of the *Isotea* is uncertain but is generally thought to be approximately middle Oligocene. Westward it grades into the mottled claystones and siltstones of the El Fausto group, which are also unconformable with the underlying strata. In western Falcón, the basal beds of the Mitare group of middle to possibly early Oligocene age transgress upon the late Eocene to possibly early Oligocene Tacal formation and date

the tectonic period as early Oligocene. After that time, western Venezuela was an unstable zone with intermittent periods of uplift and orogeny.

In the Andes of Mérida, Táchira, and Trujillo it is probable that local uplift was also taking place from early Oligocene onward, but no major unconformities have been proved. Unfortunately, very little detailed study has been made of the post-Eocene sediments and many problems concerning them remain unsolved. The post-Eocene strata consist, at least locally, of great thicknesses, in places exceeding 4,000 meters, of light blue, green, gray, and buff clays and silts mottled with red, purple, brown, and olive green, of coarse to medium sandstones, many of which are red to the lower part of the section, and of medium to coarse conglomerates. With the exception of a few marine beds, probably related to the La Rosa incursion of the northern Lake Maracaibo area, all these deposits are continental in origin. They are present along both the northwestern and southeastern flanks of the present Andes, where they have been called Betijoque group and Parángula and Rio Yuca formations. Similar sediments are found along the east flank of the Sierra de Perijá.

The Parángula and lower members of the Betijoque group are characterized by red mottling and oxidation colors. They may have been deposited in an inland basin receiving sediments from the re-elevated Cordillera on the north and northeast and from the Sierra de Perijá on the west. Low hills, the forerunners of the mighty Andes, may already have been present in central Mérida.

In this late Eocene-early Oligocene interval, all of northwestern Venezuela from approximately the Trujillo-Lara boundary northward and northwestward was involved in the earth convulsions incident to the folding and mountain-building. A land barrier was thrown across the northern end of the Maracaibo basin with the result that no marine waters could enter its lower, more southerly part, and there great thicknesses of oxidized sands, silts, and clays accumulated. Only in eastern Falcón did there remain a residue of the old Paují sea opening northeastward into the Caribbean.

At about the same time, the Cordillera de la Costa, of central and eastern Venezuela, submerged during Tinajitas time, began to rise again, though very slowly. The northern part of the Tinajitas sea was destroyed while the southern part was pushed southward. The probable connection with the Falcón basin of the late Eocene was broken. As a result, for the first time in the history of eastern Venezuela, there was formed an elongate inland basin of shallow marine deposition. This is the "eastern Venezuela" or "Orinoco" basin in which were deposited the oil-bearing sediments of the prolific oil fields of eastern Venezuela. This basin, shortly after its formation, extended from the area south of San Carlos in Cojedes eastward through the Anaco and Jusepín areas to Trinidad, where it connected with the Atlantic Ocean.

The northern shoreline of the early eastern Venezuela basin, in the region of Barcelona and eastward, is characterized by the coaly deposits of the Naricual member of the Merecure group. The initial Naricual sedimentation consists of

500 meters of carbonaceous shales, sandy shales, and sandstones (Hedberg, 1944, p. 17). This is followed by the main coal-bearing series of the Naricual mining area (González de Juana and Aguerrevere, 1938, pp. 5-32). Interbedded with the coals are sandstones and shales, about 1,000 meters in total thickness. Above the coal series is another zone of 450 meters of hard sandstones and gray shales. The Naricual formation in the Barcelona region is obviously of fresh-water to swamp to brackish-water origin and delimits, in a general way, the northern

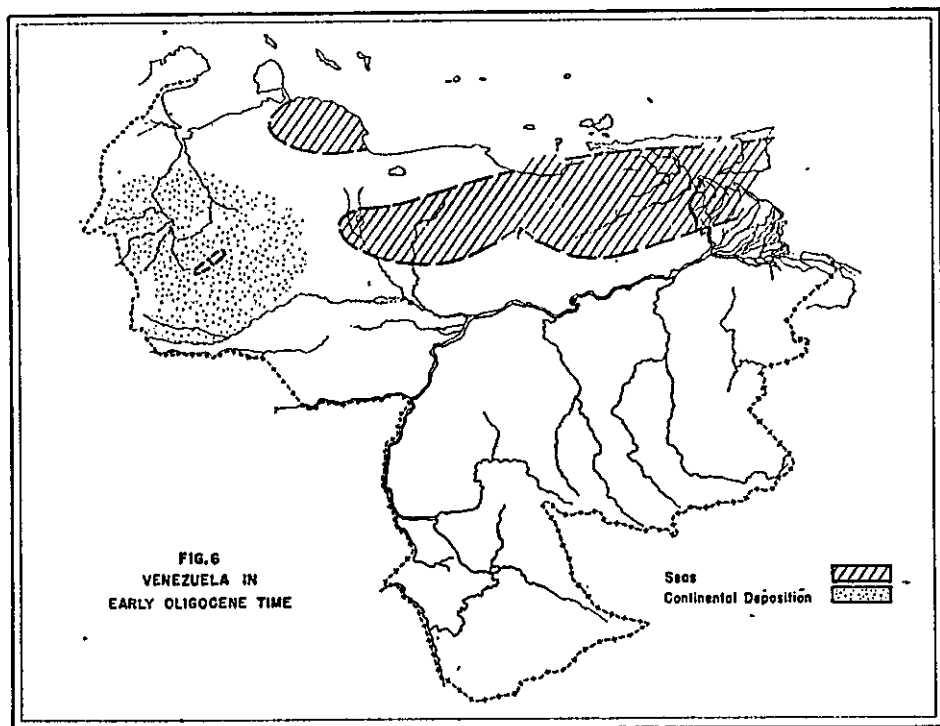


FIG. 6.—Venezuela in early Oligocene time.

strandline of the basin. Farther southeast, along the Rio Arco of northwestern Monagas, shallow marine calcareous and glauconitic beds interfinger with more typical Naricual strata. Fossils in these marine phases, while not completely conclusive, suggest very late Eocene or very early Oligocene age.

The southern shoreline of the early eastern Venezuela basin was near the Anaco area at first but, by the end of the early Oligocene, it had reached the Oficina region. At Anaco the strata of Naricual age are called the Periquito formation, while in the Oficina fields, they are known as the "U-sands" of the Oficina formation. The Periquito is composed of about 600 meters of gray, hard, massive to poorly bedded, fine- to coarse-grained sandstones and grits with inter-

beds of gray claystone, siltstone, and black carbonaceous shale (Funkhouser, Sass, and Hedberg, 1948, pp. 1870-72.)

Westward from Barcelona, the northern edge of the Naricual basin can be traced through the conglomerates and sandstones of the Fila Maestra of north-eastern Guárico and the Galera ridges of Guárico and Cojedes. Coals are again common and have been mined. West of the Rio Cojedes, the Galeras disappear beneath the llanos and their subsurface distribution is unknown. The southern edge in Guárico may be determined by the southern extent of the La Pascua formation of the Mercedes oil fields. This shows that the basin extended south of the Mercedes area.

The La Pascua formation in the Mercedes fields is composed of brackish-water to marine sandstones and dark shales with a few lignites. The thickness is about 150 meters. On the north and west, toward the axial part of the basin, the sandstones grade into dark marine shales of the Roblecito facies.

MIDDLE TO LATE TERTIARY

Once formed, the eastern Venezuela basin became the site of deposition for tremendous quantities of marine to fresh-water sediments, some localities having in excess of 6,000 meters of Tertiary strata. The north-south cross section of the basin was asymmetric, the axis being nearer the north flank than the south. The axis did not remain fixed in position, however, but migrated southward, from north of Maturín during the Oligocene to between Anaco and Oficina in the Pliocene. As most of the sediments are shallow-water in origin, it is obvious that the floor of the basin was sinking slowly throughout the Tertiary period. At the same time, the unstable mountain zone on the north was undergoing a series of uplifts. The nature of sedimentation at any place, therefore, was a reflection of the amount of submergence of the basin as compared with the amount of building-up of the depositional floor. Neither the sinking of the basin nor the rise of the mountains was constant, it sometimes being less, sometimes more, and sometimes stopped altogether.

Following Naricual time, further elevation of the northern mountains with its resultant rejuvenation of streams and increased erosion caused a greater distribution of northerly derived sediments than formerly. This process was further aided by a sinking of the basin and southward shift of the axis. The change brought about by these conditions can be seen clearly in the Anaco area, where the clean Periquito sandstones whose components were brought from the south are overlain by softer sandstones of more varied composition and northern derivation. This lithologic transformation marks the contact between the Periquito and Oficina formations. On the north, a somewhat similar lithologic change, although less obvious, is believed to be the commencement of the post-Naricual, or Santa Inés, cycle of deposition.

With deepening of the basin and more stable conditions in the north, the early Santa Inés sea gradually began to invade eastern Venezuela from the east. The

characteristic deposit of the axial part of this sea is the Carapita formation, composed almost entirely of dark gray or brownish gray foraminiferal shales, laid down in a moderately deep open-water environment. The Carapita shales are found in the foothills and wells of northeastern Anzoátegui and northern Monagas. Westward toward Barcelona the Carapita grades into shallower-water shales and sandstones of the Capiricual formation but eastward it thickens at the expense of other members of the Santa Inés group and attains a thickness of more than 2,000 meters. The subsurface equivalent of the Carapita formation in central and southern Anzoátegui and Monagas is the Oficina formation, which includes the main oil-producing zones of the Anaco, Oficina, Temblador, and Tucupita areas. In the Oficina field, the formation consists "dominantly of alternations of gray, dark gray, and brownish gray shales, interlaminated and interbedded with light gray fine- to coarse-grained sandstones and siltstones. Thin lignites and lignitic shales, green and light gray claystones with siderite spherules, sideritic glauconitic sandstones, and thin cone-in-cone limestones are important minor constituents" (Hedberg, Sass, and Funkhouser, 1947, p. 2107). The thickness is 600-1,200 meters. The Oficina of the Anaco fields differs in having a greater predominance of shale and a lesser quantity of coarse gritty sandstones and glauconitic fossiliferous sandstones (Funkhouser, Sass, and Hedberg, 1948, p. 1869) while the section is much thicker, being some 2,250-3,000 meters. The character of the deposits in the Oficina field shows that that area lay not far from the shoreline throughout most of Oficina time and that the position of the strand fluctuated north and south of the field. The most southerly advance of the shoreline took place at about middle Oficina time (late middle to early late Oligocene) when the most marine phases were deposited in the Anaco area. At its greatest extent, the Oficina sea must have approached the present course of the Orinoco River.

Southwestward in Guárico, the equivalent of part of the Carapita-Oficina is found in the Roblecito formation of the Mercedes oil fields. The Roblecito is a dark shale facies interfingering southward and southeastward with sandstones, some of which are important reservoirs for oil.

In late Carapita-Oficina time, uplift with perhaps some folding and thrusting was renewed in the northern mountain area. The waters covering the basin shallowed and non-marine sediments were distributed over wide areas formerly occupied by the Carapita sea. In the extreme north, Carapita strata were themselves uplifted and eroded so that the succeeding La Pica deposits lie with marked unconformity on them. This unconformity dies out southward but reappears on the structural high of the Anaco trend, which began to form at about this time. It is not to be expected; however, that the change took place at exactly the same time throughout the basin. The first effects were felt in the west perhaps as early as middle Oligocene; there the Roblecito shales change upward into Chaguaramas brackish-water shales and sandstones and fresh-water clays. Later, in northeastern Anzoátegui, the Carapita shales were supplanted in the late Oligocene by

chert-pebble conglomerates, sandstones, and silty shales of the Uchirito formation, but marine shales of the Carapito facies continued to be deposited in northern Monagas until the early Miocene. At Anaco, the middle, more marine members, of the Oficina formation were replaced by the upper, brackish to occasionally fresh-water members.

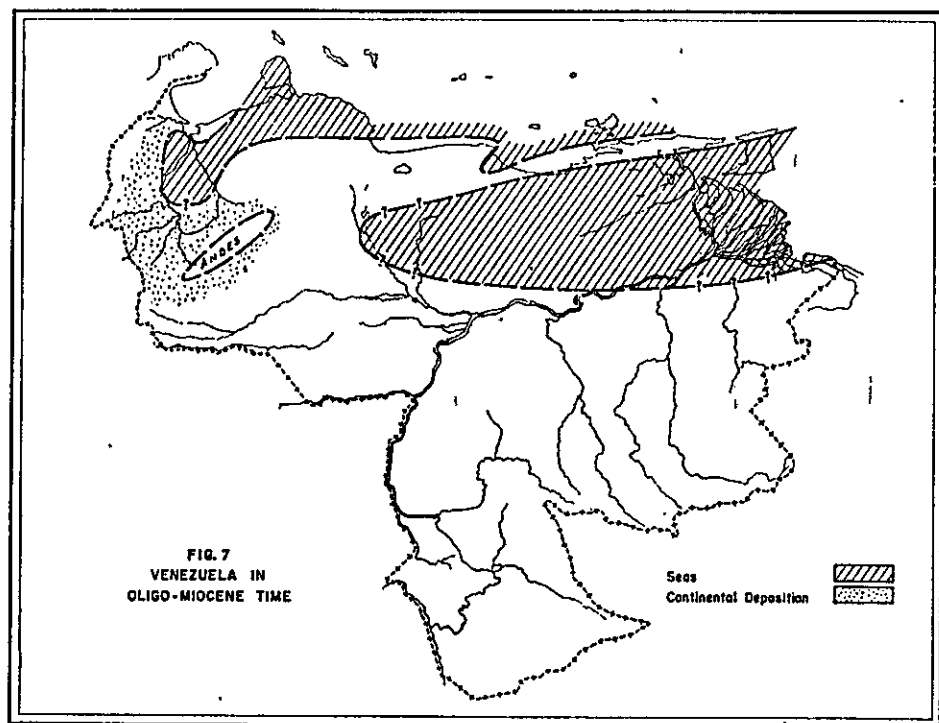


FIG. 7.—Venezuela in Oligo-Miocene time.

Before the end of the early Miocene, submergence again dominated filling. The new shallow sea had as its principal deposit greenish gray, glauconitic sands, greenish gray clays and sandy clays (Freites formation). The axis of deposition now lay close to the Anaco area while the southern shoreline was at its farthest extent, probably well south of the Rio Orinoco. The marine La Pica formation, the important oil producer on the Santa Bárbara-Jusepín trend and possibly at Pedernales, is the Freites equivalent along the northern rim of the basin. Accumulation of oil has taken place along the La Pica shoreline. The basin did not have as great a westward development as in Carapita time but the exact boundary is not known.

Marine inundation of the Maracaibo basin during the middle to late Tertiary, while geologically of little significance, is of extreme importance to the petroleum industry. It will be recalled that, due to late Eocene-early Oligocene

tectonic movements, a land barrier in the northern Maracaibo area prevented marine waters from entering any part of the basin. Continental sedimentation of the Betijoque-Parángula type continued for a considerable time. Then, at some time during the late Oligocene the Falconian sea broke across the barrier and a shallow marine basin was again established. The resultant marine strata, called the La Rosa and Lagunillas formations in the Bolívar Coastal fields, are the principal producing formations of western Venezuela. They grade shoreward into the mottled Betijoque, Guayabo, and El Fausto groups all of which contain some very thin brackish to shallow marine beds interfingering with the predominant continental beds.

The La Rosa has a thickness of 180-250 meters and consists of interbedded soft gray and grayish green clay shales, sandy shales, and sandstones. The formation thins and interfingers with the brackish-water deposits in all directions from the Bolívar Coastal area except toward the northeast, where, at Quirós, in the southeastern part of the District of Miranda, Zulia, there are about 1,000 meters of La Rosa strata. From Quirós, the La Rosa sea opened into Falcón where the Cerro Pelado and Socorro formations were being deposited. The Lagunillas formation is composed chiefly of interbedded gray, brown, and greenish gray shales, mottled clays, soft poorly consolidated sandstones, and lignites. The environment of deposition was one of rapid changes from brackish-water to non-marine to marine to fresh-water (Sutton, 1946, p. 1698), and probably represents the retreating phase of the La Rosa sea. The Lagunillas formation has a maximum thickness of 450 meters in the Bolívar Coastal fields but thickens westward to about 900 meters in the District of Urdaneta, Zulia. Still farther westward, the Lagunillas grades into the fresh- to brackish-water Los Ranchos formation, which consists of light green and light gray clays with some red mottling, sandstones, and conglomerates.

The final disappearance of the La Rosa sea probably coincided with important uplifts in the Andean zone, for it was near this time (middle to late Miocene) that the mottled Parángula and Betijoque facies gave way to the extremely coarse sediments of the Rio Yuca and Sanalejos formations. The Rio Yuca beds are characterized by light bluish, greenish, yellowish, and buff colors. They contain innumerable thick intercalations of poorly consolidated boulder conglomerates made up in large part of pieces of crystalline igneous and metamorphic rocks. There seems little question that the Rio Yuca and Sanalejos conglomerates were derived from an uplifted Andes where erosion had already stripped away the Cretaceous and Tertiary strata capping the crystalline core. Actually the evidence suggests that erosion had cut down to the metamorphics even before this, for the Isnotú formation, which is the middle member of the Betijoque group and directly underlies the Sanalejos, contains a heavy-mineral suite with metamorphic components, such as andalusite, sillimanite, zoisite, kyanite, biotite, muscovite, and glaucophane (Sutton, 1946, p. 1706). Nevertheless, the Andes had not yet assumed their present-day dimensions for there is no evidence for

unconformity or break in deposition between the Isnotú and Sanalejos or Parángula and Rio Yuca except in the northern Andes in the vicinity of the Mene Grande field or north of Barinitas.

That orogenic movements continued throughout the time of deposition of the Rio Yuca formation is shown by the repetition of coarse conglomerates within that formation. Moreover, the conglomerates become coarser upward in both the Rio Yuca and Sanalejos formations, indicating that the mountain elevations were constantly increasing. Finally, in the Pliocene, the culmination of the Andean movements was reached, with great folding, thrusting, and faulting of the strata. Not only was the central part of the Andes affected as formerly; but also the present-day flank areas. All the rocks, from pre-Cambrian to Pliocene, were folded and deformed in a zone probably somewhat more extensive than that of the present Andes. Equally important movements simultaneously affected the Sierra de Perijá.

The more central parts of the Maracaibo basin were relatively undisturbed by the Andean orogeny. It is possible that the basin acted as a sort of bastion against which the Mérida Andes were thrusting themselves from the southeast and the Sierra de Perijá from the west (De Cizancourt, 1933, pp. 220-25). Whether deposition continued uninterruptedly in the deeper parts of the basin during the Andean orogeny remains a question, since these areas are at present under the waters of Lake Maracaibo. Wherever the relationship between the Pleistocene formations and the underlying strata can be seen, however, it is one of pronounced unconformity.

The late Miocene-Pliocene orogeny was not limited to western Venezuela. Important disturbances took place in all the mountain areas across the country and these were accompanied by general shallowing and shifting of axes in the basins. The greatest part of the deformation now visible in the Serranía del Interior of eastern Venezuela and the foothills of central Venezuela originated at this time. In the north, the Santa Inés, La Pica, Chaguaramas, and older formations were all intensely folded and thrust southward, resulting in the very complicated structures of the Serranía del Interior and the great zones of folding and thrusting in the foothill and Galera belts of Guárico and Cojedes. This deformation died out southward but its effects can be seen over great distances toward the south in the superficial crumpling of the soft middle to late Tertiary strata. The Anaco and similar trends were formed by these movements although they may have begun their history somewhat earlier. Much of the faulting of the southern flank of the eastern Venezuela basin is related to this period, although in many places the displacements seem to have taken place along earlier established trends.

From the point of view of the oil industry, the importance of the late Miocene-Pliocene orogeny can not be overstressed. Practically every structure on which oil is found in Venezuela, whether it be of folding, thrusting, or simple normal faulting, or whether it be located in the west, center, or east, either originated or

received its principal configuration in the movements of that period.

The Mio-Pliocene movements are also responsible for the development of the large geographic features of the country. The Andes and Sierra de Perijá attained their maximum elevations and extension. The Lara-Falcón ranges rose. The Cordillera de la Costa and Serranía del Interior were re-elevated, reaching greater dimensions than they had before. The foothill zones and some outlying structural highs, such as the Anaco trend, were either originated or further developed. The basin areas, however, remained at somewhat lower levels than they are to-day, the present configuration of the llanos being a slightly later development.

One of the most complete sections of marine Oligocene and Miocene strata known from the Caribbean area is to be found in Falcón. Nevertheless, there are many unconformities present and the beds, on the whole, were deposited in shallow water. There is some evidence to indicate that during the Oligocene the Falconian sea was restricted to a narrow channel cutting across Falcón in a northeasterly direction and deepening toward the northeast (González de Juana, 1938, pp. 132-38). Later, however, the sea was more widespread, though still shallow.

At some time during the mid-Tertiary, minor igneous activity affected southern Falcón and northern Lara when dioritic intrusions in the form of dikés cut through the sediments. The youngest intruded rocks belong to the middle Oligocene San Luis formation, but the time of igneous activity can not be dated precisely.

LATE TERTIARY-QUATERNARY

While Mio-Pliocene orogeny was taking place in the north, the axis of the eastern Venezuela basin was shifting southward, reaching the Anaco area in the Pliocene (Funkhouser, Sass, Hedberg, 1948, p. 1883). At the same time, the basin shallowed and the Freites marine sediments were replaced by fresh-water to brackish deposits of the Sacacual group. On the north, along the edge of the foothills, the Sacacual beds lie with marked unconformity upon older strata, as might be expected. Southward the unconformity disappears and there is apparent transition from La Píca or Freites to Sacacual. The unconformity is again present on the Anaco high which existed as a northeasterly striking ridge rising above the low llanos surface surrounding it. Enough of the structure was exposed so that erosion stripped away the complete section of Freites formation that had been deposited upon its crest. As time progressed, erosion lowered the ridge while widespread distribution of northerly derived sediments raised the level of the plain until it was built up to the ridge level, after which time Sacacual sedimentation spread out over the ridge hiding it from view. Farther south, Freites-Sacacual (Las Piedras) relationships are again transitional.

A very important phase of Sacacual deposition occurred in northern Monagas along the foot of the Serranía. Here coarse sands, gravels, and clays were dumped by mountain streams upon the upturned edges of oil-bearing Cretaceous and

Tertiary strata. Subsequently, the oil escaped from its source beds into the porous Sacacual reservoir sediments, resulting in the great accumulation of the Quiriquire field. The Sacacual beds of this area are known as the Quiriquire formation.

Renewal of tectonic disturbances caused folding of Sacacual beds along the mountain front at the end of the Pliocene. At the same time there was a general uplifting and eastward tilting of the basin (Funkhouser, Sass, and Hedberg, 1948, p. 1883).

Following the main Andean orogeny of Mio-Pliocene time, the great mountain chains were subjected to erosion. As a result, large areas of alluvial gravels, sands, and clays were deposited all along the foot of the mountains, in the stream valleys within the mountains, and in the surrounding basin regions. These deposits are roughly of Pleistocene age and are known by a variety of local formational names, such as El Milagro, Necesidad, Carvajal, and Guanapa in the west and Mesa in the east. The well known ferruginous clay and cross-bedded sand that underlie most of the city of Maracaibo and the sands, clays, and gravels that form the Mesas of Guárico, Anzoátegui, and Monagas belong to these strata. Presumably there are marine strata within Lake Maracaibo that were laid down contemporaneously with the Pleistocene continental beds. In general, the thickness of the continental deposits does not exceed 150 meters.

Uplift did not terminate with the Pliocene, though later movements are of minor importance. As a consequence, the Pleistocene alluvial deposits have been elevated and exposed to erosion and this has resulted in the present-day topography of terraces, mesas, and sea cliffs.

To-day the processes of erosion and sedimentation continue: erosion of the mountains, the hills, the mesas, and the terraces; sedimentation in the valleys, the llanos, the lakes, and the sea. Lake Maracaibo is once again under shallow water but the sea has been drained from the eastern Venezuela basin and pushed back still farther by the ever-growing Delta Amacuro. Occasional earthquakes warn that the mountains are not yet stable and may again be thrown into disturbances that will change the configuration of the country and bring about a new chapter in the geological history of Venezuela.

PART II—OIL FIELDS OF VENEZUELA

WESTERN VENEZUELA

1. MARACAIBO-MARA FIELDS

H. J. FICHTER AND H. H. RENZI¹¹

The oil fields west of the city of Maracaibo made a modest contribution to the production of Venezuela during the period from 1925 to 1941. Since that time their output has greatly increased, because of the discovery of prolific deeper

¹¹ Shell-Caribbean Petroleum Company.

pools, and amounts now to about one sixth of the current production of all Venezuela. The cumulative total production of these fields is approaching 300 million barrels, the greater part of which came from the La Paz field.

The oil produced in this area is generally of good quality, with average gravities ranging from 29° to 35° A.P.I. and a gasoline content of about 25 per cent. The Boscan field is an exception, however, inasmuch as it furnishes very heavy oil, of 11° A.P.I., which is so viscous as to be difficult to pump through a pipeline.

The stratigraphic sequence as known from the general Maracaibo-Mara area is summarized in the stratigraphic column (Fig. 10). The Cretaceous, developed in a marine limestone and shale facies (Cogollo-La Luna-Colon limestone and Colon-La Paz shales, respectively), rests with a pronounced unconformity on the pre-Cretaceous basement. The latter consists, in the Maracaibo region, mainly of granodioritic igneous rocks. However, toward the Sierra de Perijá, the basement is much more complex and includes also Jurassic-Triassic sediments of the redbed type (La Quinta formation). The Paleocene is represented by the fossiliferous limestones and sandstones of the Guasare formation, the Cretaceous-Tertiary contact being essentially transitional. The Eocene Concepción formation, consisting mainly of a thick coal-bearing shale-sandstone sequence, overlies the Guasare with a slight unconformity. It is unconformably overlain by a predominantly shaly-clayey series (Los Ranchos-La Villa formation) of post-Oligocene age which is usually referred to as the Younger Tertiary and which consists, to a considerable extent, of mottled clays. Toward the Perijá, this Younger Tertiary transgression also truncates the Cretaceous and eventually reaches down into the pre-Cretaceous.

The first oil field to be discovered in the region of Maracaibo Lake was the Totumo field, 90 kilometers west of Maracaibo, at the edge of the Perijá Mountains, where there are large oil seeps. The first well was drilled in 1914 and oil was found at the unconformity between the La Quinta formation and the overlapping formations of Tertiary age. Initial production was high in some wells but it declined rapidly. About a dozen wells were drilled but no sustained production was developed. The field is abandoned.

La Paz was the second field discovered and is also the most important in this region (Figs. 9 and 11). The first well was drilled in 1922, by the Venezuelan Oil Concessions Limited (Shell subsidiary), relying on the presence of large oil seeps and an anticlinal structure which was mapped at the surface. In the early days, production came mostly from sandstones and limestones of the Guasare formation, the remainder from sandstones of the Concepción formation.

The field was slowly declining when in 1944 it was decided to drill deeper, down to the Cretaceous limestone which has good oil showings at its outcrop along the foot of the Perijá Mountains, and which was known to yield oil in the Barco concession in Colombia, 240 kilometers southwest of La Paz. The result of the first test was encouraging. Subsequent deeper wells came in with an

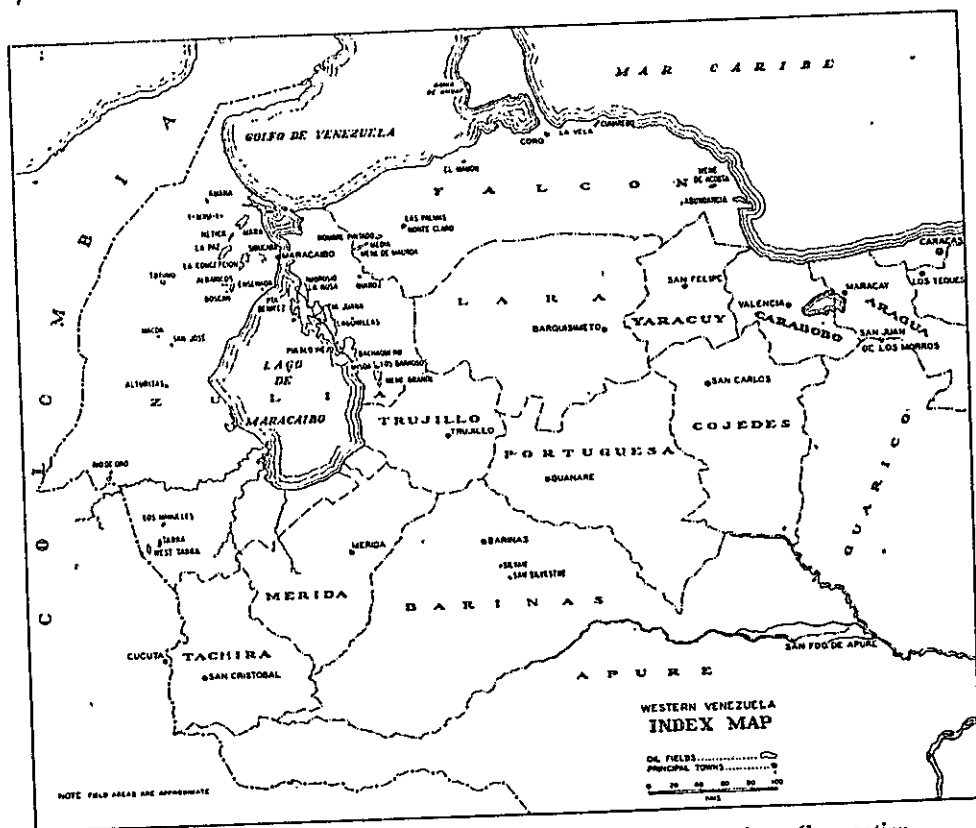


FIG. 8A.—Index map: Western Venezuela. Courtesy of Creole Petroleum Corporation.

average initial production of about 5,000 barrels per day, as compared with about 500 barrels for the shallow wells, and the output of the La Paz field increased more than 100-fold in the 6 years from 1942 to 1948 and is increasing still.

The reason for the large oil production from the Cretaceous limestone here appears to be complex. The limestone is dense, with an average porosity of 1 or 2 per cent, but the thickness of the productive section (more than 500 meters) seems to make up for the low porosity. Also the productivity varies very much from one well to the next, indicating that the more prolific wells are fed from fissures rather than from the primary porosity of the limestone. Another essential factor is the outstanding size and height of the La Paz structure (Figs. 9 and 11).

The Concepción field was discovered in 1924, one year after La Paz, and it is of similar type. There is a complicated faulted anticline at the surface, but without seeps (Fig. 12). Until recently, oil was producing exclusively from several sands in the Concepción formation (Eocene). Particularly, the Ramillete sand gave a high yield, especially on the Ramillete structure (Fig. 12, left half), where many wells had initial production of 1,000 barrels per day.

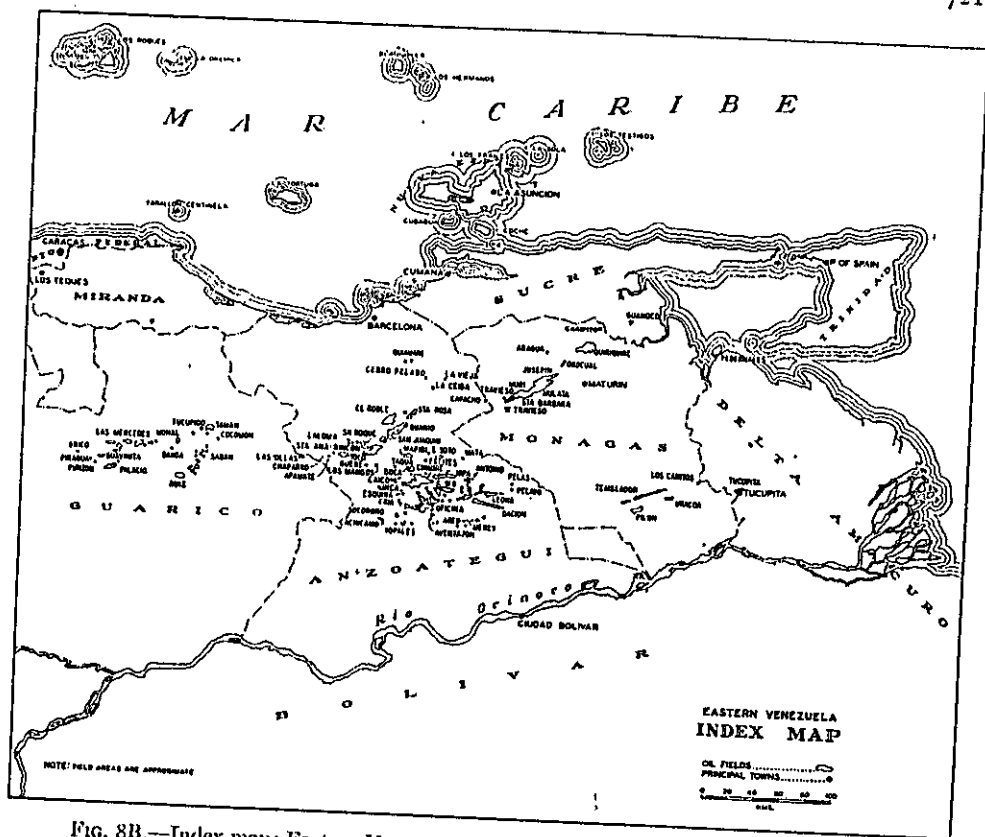


FIG. 8B.—Index map: Eastern Venezuela. Courtesy of Creole Petroleum Corporation.

The Concepción field reached its peak production in 1931, with a daily average of 12,500 barrels, or about 200 barrels per well. More wells were drilled but the field declined and in 1947 the daily yield per well had decreased to 30 barrels. A single deep well was drilled then to the Cretaceous limestone and this alone at present is producing as much oil as all shallow Eocene wells together.

The successful development of production from the Concepción and Guasare formations at La Paz and La Concepción in the early twenties led to a further search for similar accumulations. In 1928, the Venezuelan Gulf Oil Company (now known as the Mene Grande Oil Company) discovered the Amana field which has produced only a small quantity of oil from the Concepción formation, on a faulted structure, 40 kilometers northwest of La Paz. In 1929, the Orinoco Oil Company followed with the Netick field, in the northern prolongation of the La Paz structure, but obtained only an unimportant quantity of oil.

A more important discovery was made again in 1947, when the Richmond Exploration Company opened the Boscán field in the southern prolongation of the Concepción field and produced heavy oil from the Concepción formation

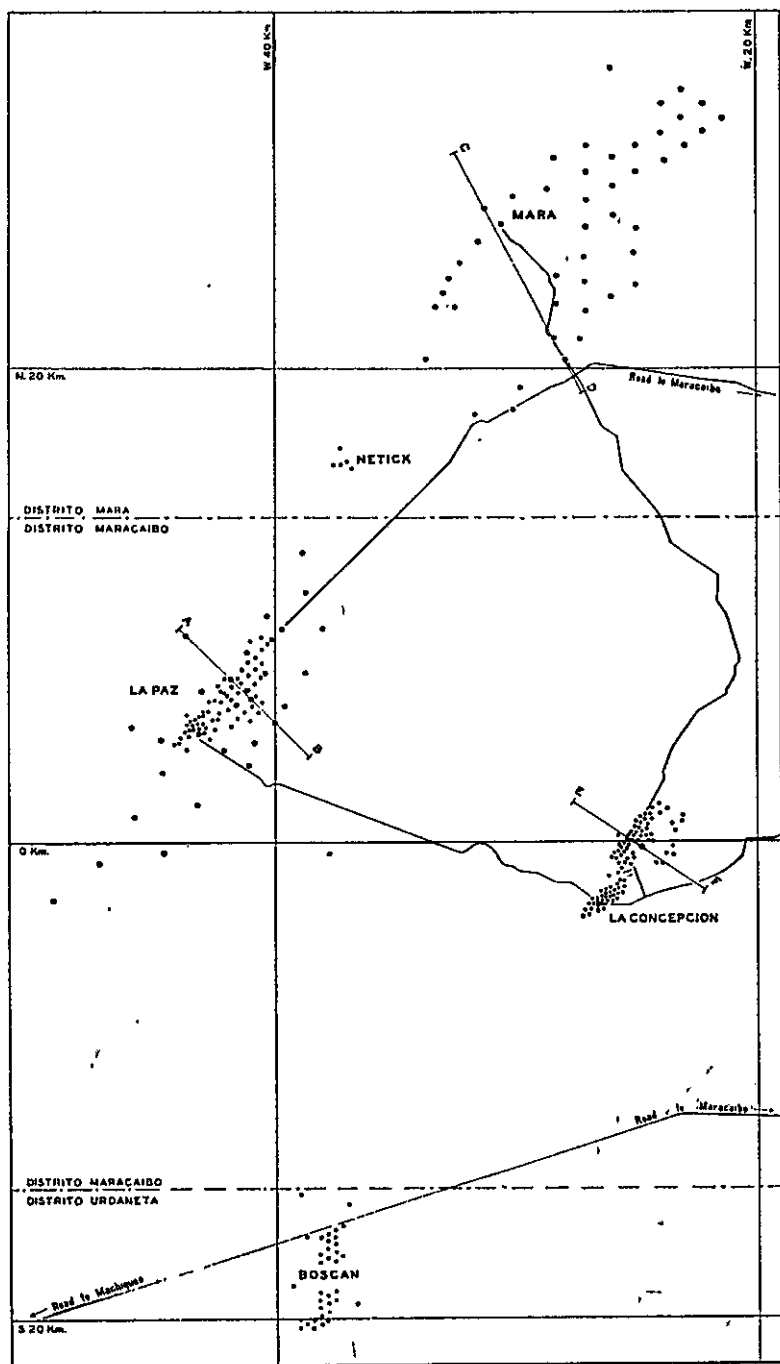


FIG. 9.—Location map: La Paz, Netick, La Concepción, and Boscan fields.

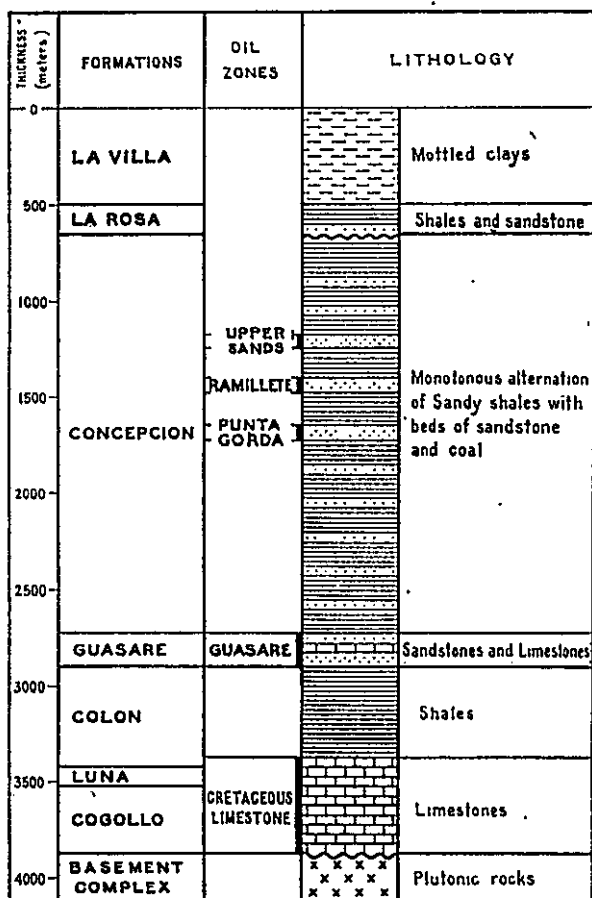


FIG. 10.—Stratigraphic column: Maracaibo-Mara region.

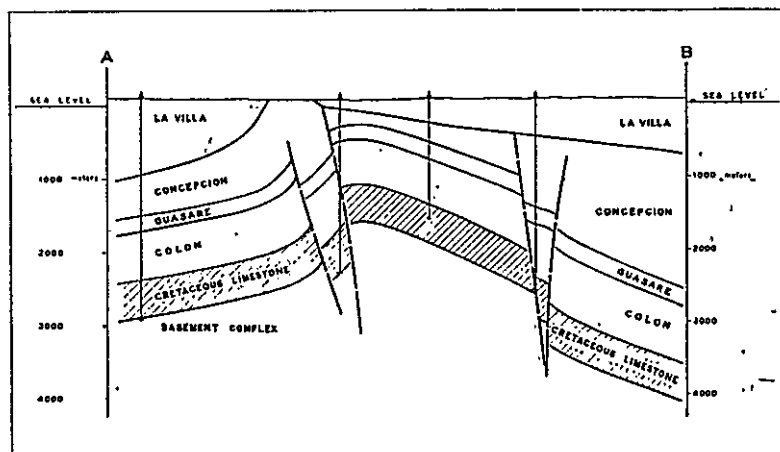


FIG. 11.—Cross section: La Paz field.

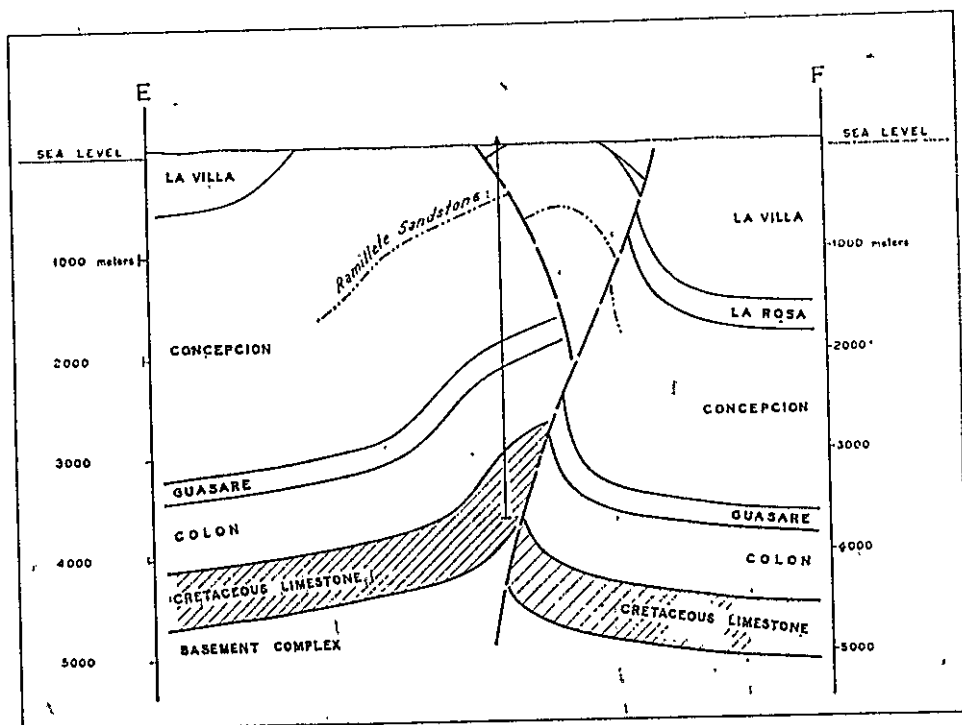


FIG. 12.—Cross section: La Concepción field.

(Fig. 10). Subsequently Richmond obtained light oil from the same formation at La Ensenada, from a single well 20 kilometers south of Maracaibo.

Since 1944, the main interest has been in the prolific production from the Cretaceous limestone which had been discovered at La Paz, and the most important outcome has been the development of the Mara field (Figs. 9 and 13).

Mara was discovered by the Caribbean Petroleum Company (Shell subsidiary) in 1945, relying on a combination of geological and geophysical methods, and after earlier tests for shallow production had failed. The oldest parts of the Mara field originally were called Los Tetones and Kilometer 24 before it was realized that they belong to one much larger field which is held by four different companies. To-day, Mara is one of the important oil fields of Venezuela and ranks second only to La Paz as a Cretaceous limestone field.

Similar limestone oil was found again in 1948, at Sibucara, by the Venezuelan Oil Concessions Limited (Shell subsidiary). A single well so far has been completed, 4 kilometers west of the airport of Maracaibo, but this well alone currently produces as much oil as the whole Concepción field ever produced in its early days.

More detailed data on the Concepción and La Paz fields have been published by the staff of the Caribbean Petroleum Company (1948).

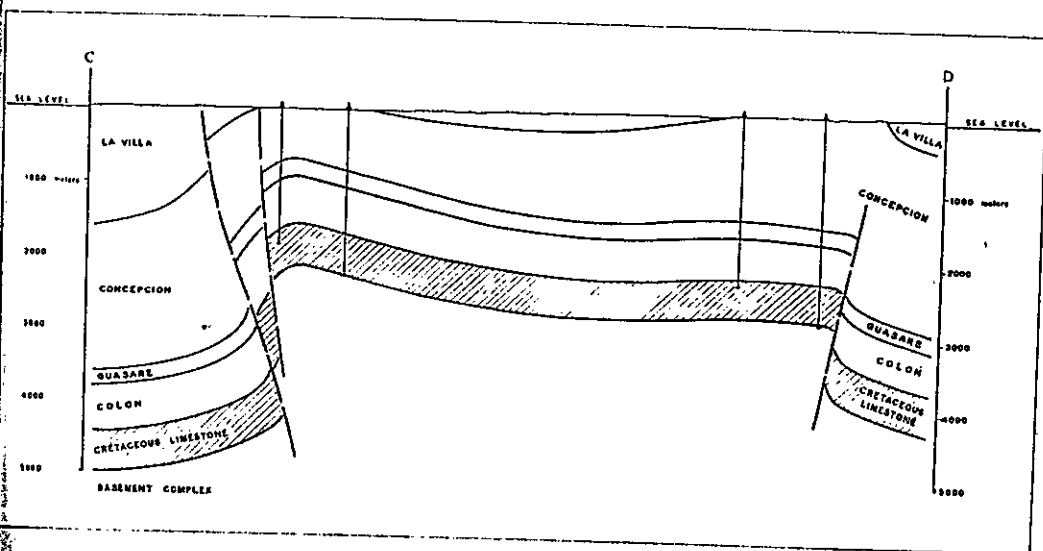


FIG. 13.—Cross section: Mara field.

2. MENE GRANDE FIELD

H. J. FICHTER AND H. H. RENZI¹²

The primitive method of drilling on seeps was responsible for the discovery of Mene Grande, one of the most prolific oil fields of Venezuela, whose cumulative production at present is nearly 350 million barrels and which still contributes about 3 per cent of the current production of all Venezuela.

The field is 15 kilometers inland on the east side of Lake Maracaibo, on the highway from Maracaibo to the Andes. It owes its name to the large seeps of asphalt which may be seen from the highway, on the hill north of the office building (Fig. 14). It was discovered in 1914, by the Caribbean Petroleum Company (Shell subsidiary), and is therefore the oldest commercial oil field in Venezuela.

The oil found at Mene Grande is of several different qualities, heavy in the younger formations and lighter in the deeper levels. The average gravity is about 20° A.P.I. and the average gasoline content about 12 per cent. The oil is transported through a pipeline to the shore of the Lake where most of it is processed, in the old refinery of San Lorenzo.

The stratigraphic sequence (Fig. 15) begins with the Eocene represented essentially by a lower sandstone series (Misoa sandstone) and an upper shale sequence (Pauji formation). The uppermost formation preserved below the truncating Younger Tertiary unconformity is the Mene Grande formation, probably Claiborne (upper middle Eocene) in age. Thus, upper Eocene and Oligocene are

¹² Shell-Caribbean Petroleum Company.

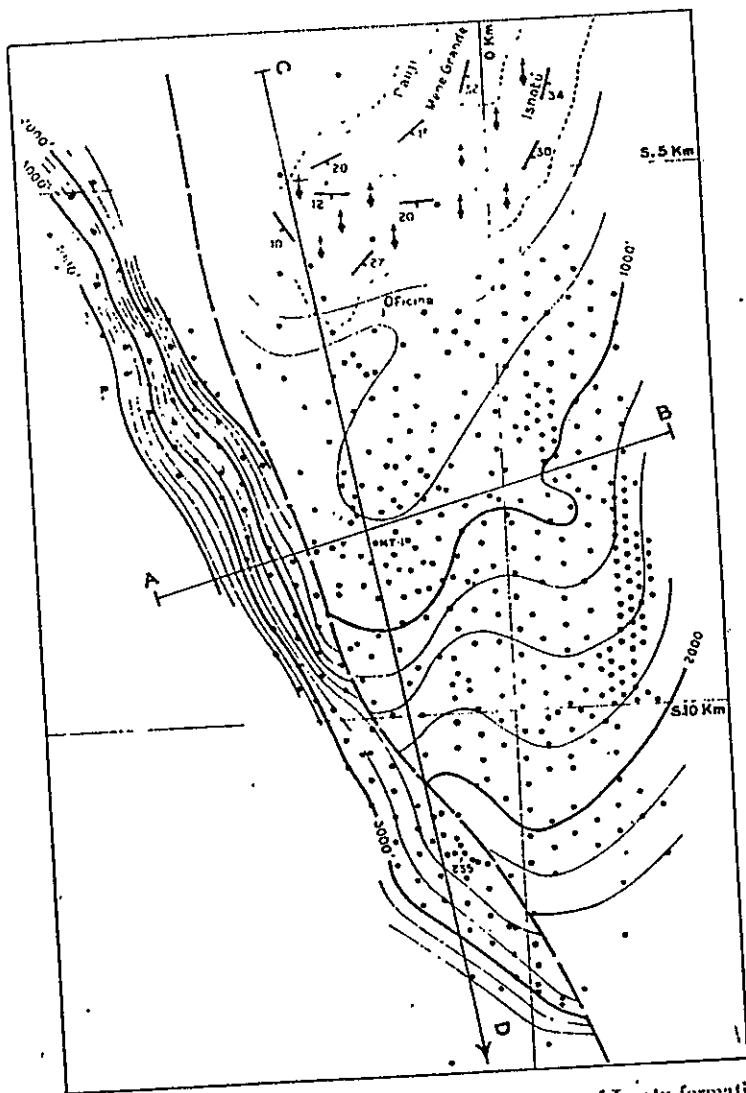


FIG. 14.—Mene Grande field: structural contours on top of Isnotu formation. Arrow-head symbols indicate asphalt seeps.

missing. On the highest parts of the Mene Grande structure, the Miocene unconformity cuts down almost into the Misoa sandstone. A minor unconformity within the Younger Tertiary itself separates the Isnotú and Sanalejos formations. Exploration of the lower Eocene-Cretaceous levels in the Mene Grande field has not yet been undertaken on account of the considerable depth involved. The oil first discovered at Mene Grande came from unconsolidated sands of the Isnotú formation. As may be seen in Figure 14, this oil is accumulated on a

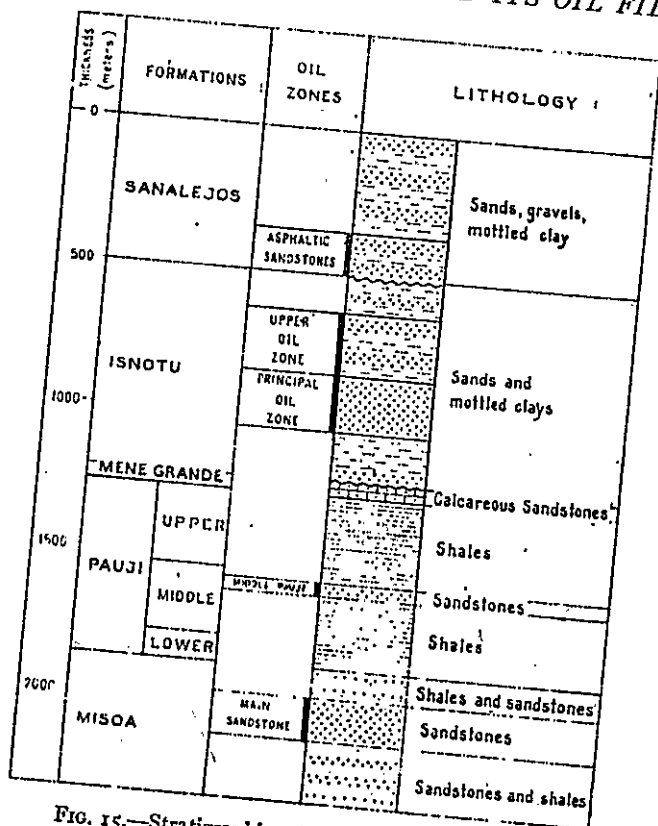


FIG. 15.—Stratigraphic column: Mene Grande field.

southward plunging anticline, without structural closure on the north. It appears therefore that the oil is kept in the reservoir by nothing more than the asphalt seepages plugging the outcrop.

In 1925 it was found that oil is present also in the Eocene formations below the unconformity, and systematic drilling eventually led to the geological picture which is shown in Figures 16, 17, and 18.

A part of the Eocene production comes from the middle Pauji sandstone. The trap is strictly stratigraphic, the sandstone being cut off by the angular unconformity.

Another kind of Eocene accumulation was discovered in 1930, with the MT-1 well which tapped a dome in the Main sandstone of the Misoa formation. This well has produced nearly 15 million barrels of oil, apparently because of its position at the top of a closed dome. It is curious that this sandstone actually is light and nearly impermeable so that the oil apparently comes essentially from the sandstone.

Prolific wells such as MT-1 raised high hopes for the Eocene possibilities of the Mene Grande field. By drilling and through seismic investigations, two more

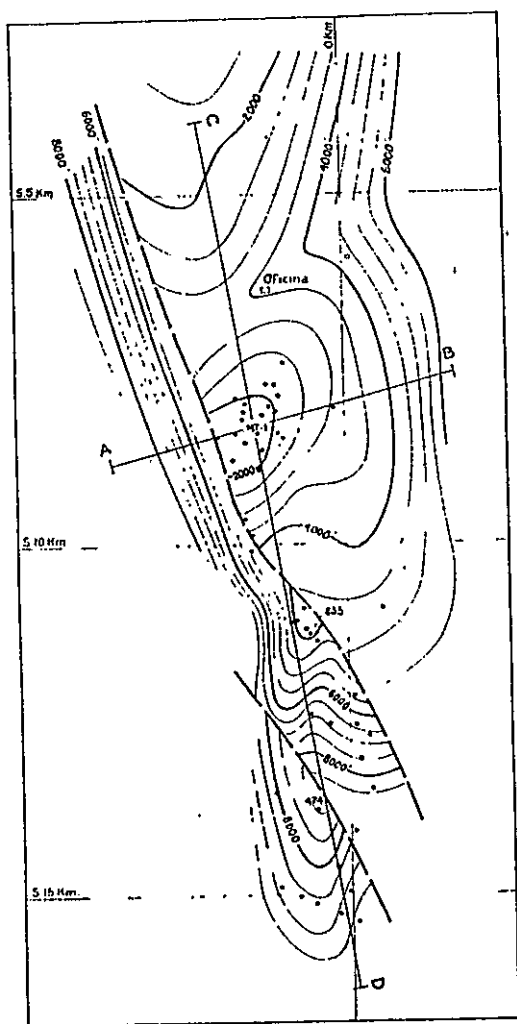


FIG. 16.—Mene Grande field: structural contours on top of Misoa formation.

closed structures were found in the Eocene, farther down the south plunge (Fig. 18), but the results were somewhat disappointing, for the greater part of the oil is still produced from the Isnotú formation.

Early development of the Isnotú production had concentrated on the shallow northern and eastern parts of the structure. In more recent years, drilling has been concentrated on the down-faulted west flank, where the wells have to be drilled much deeper but where also the yield per well is greater and the oil is of somewhat better grade. The Isnotú formation of the west flank is structurally closed by the main fault. Production is governed also by porosity, however, since the sands are better developed farther down on the flank.

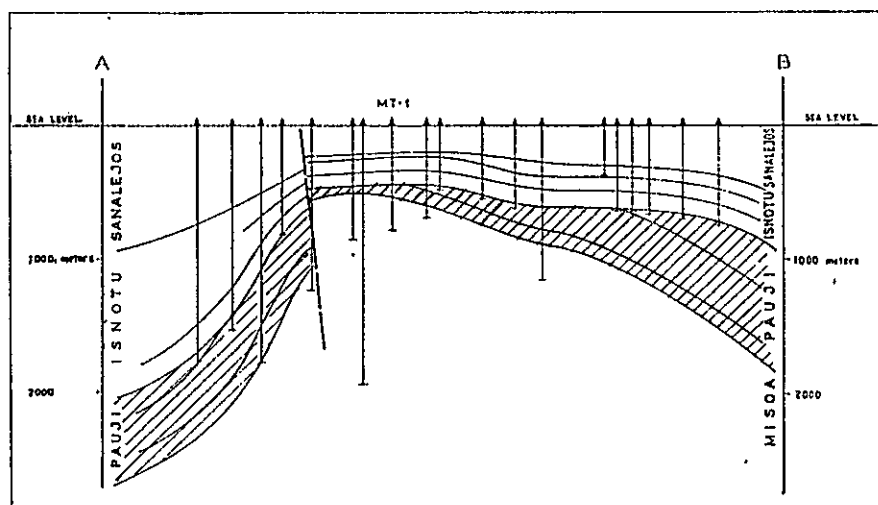


FIG. 17.—Cross section: Mene Grande field.

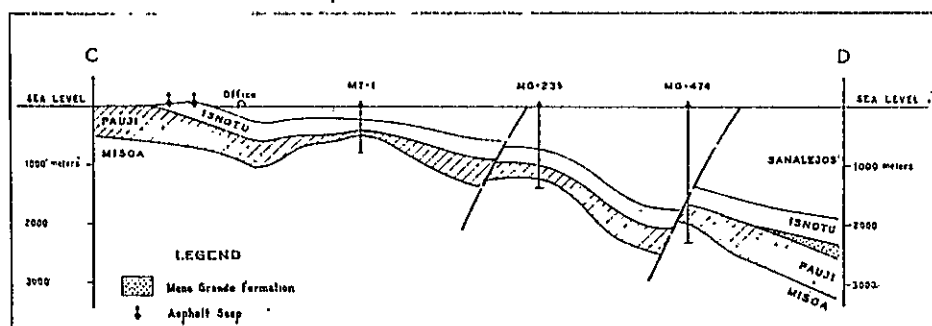


FIG. 18.—Longitudinal section: Mene Grande field.

Further data on the Mene Grande field may be found in a publication by the staff of the Caribbean Petroleum Company (1948).

3. TARRA FIELDS

H. J. FICHTER AND H. H. RENZI¹³

The Tarra group of oil fields is near Rio Tarra, about 100 kilometers southwest of Lake Maracaibo, close to the Colombian border. There are three individual fields: Las Cruces, Los Manueles, and West Tarra.

These are small fields, as compared with others in the Maracaibo basin, since their present production amounts to only about 1 per cent of the current production of Venezuela. Nevertheless, they have been producing for more than 20 years and their cumulative production has reached 115 million barrels.

¹³ Shell-Caribbean Petroleum Company.

The oil produced from the Tarra fields is of good quality, having an average gravity of about 30° A.P.I. and a gasoline content of about 20 per cent.

The stratigraphy of the general Tarra area is shown in the columnar section (Fig. 19). A major unconformity separates the Cretaceous from the pre-Creta-

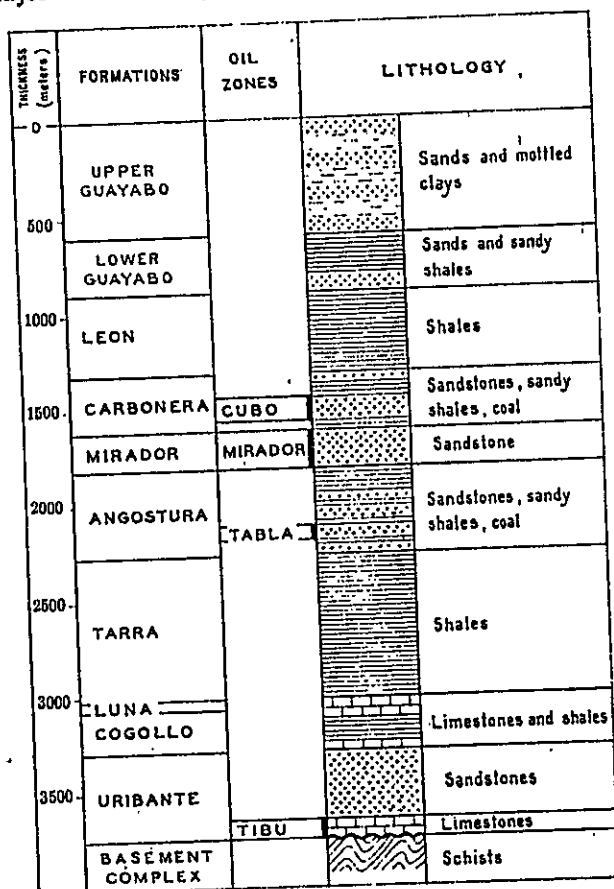


FIG. 19.—Stratigraphic column: Tarra fields.

ceous basement. The Cretaceous, compared with that of the Maracaibo region, shows a somewhat different facies development: part of the Cogollo limestone is developed in shales, and a sandstone member (Aguardiente member of the Uribante or Tomón formation) attains considerable thickness. The Paleocene Guasare formation, as such, is missing. Its age equivalent is represented by the uppermost part of the Tarra shales. The marine Tarra shales are conformably overlain by a thick coal-bearing sequence of shales and sandstones mainly of a brackish-water origin, subdivisible into Angostura, Mirador, Carbonera, Leon,

and Guayabo formations. This sequence represents, so far as is known, the Eocene, Oligocene, and Miocene. In contrast to the northern part of the Maracaibo basin, sedimentation seems to have been essentially continuous throughout early and late Tertiary time.

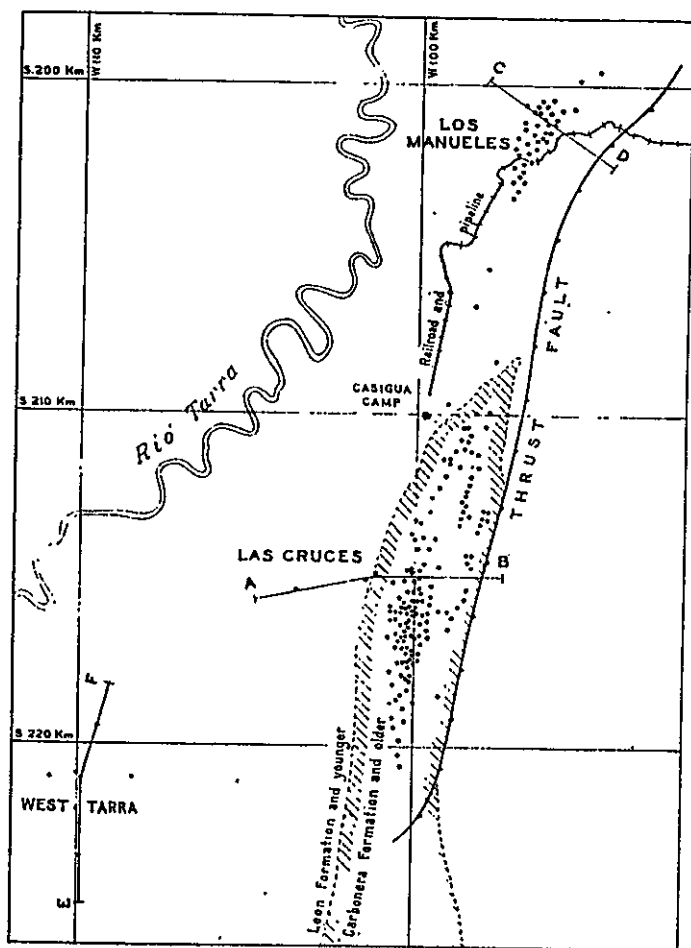


FIG. 20.—Tarra fields.

The Cruces field is the oldest of the Tarra fields. It was discovered by the Colon Development Company (Shell subsidiary) in 1916, at which time the area was still inhabited by the Motilone Indians. The discovery was guided by the occurrence of oil seeps and by the presence of an anticlinal structure which may be seen from far away, rising out of the swampy plains (Figs. 20 and 21).

Further surface geological work indicated the presence of a second dome,

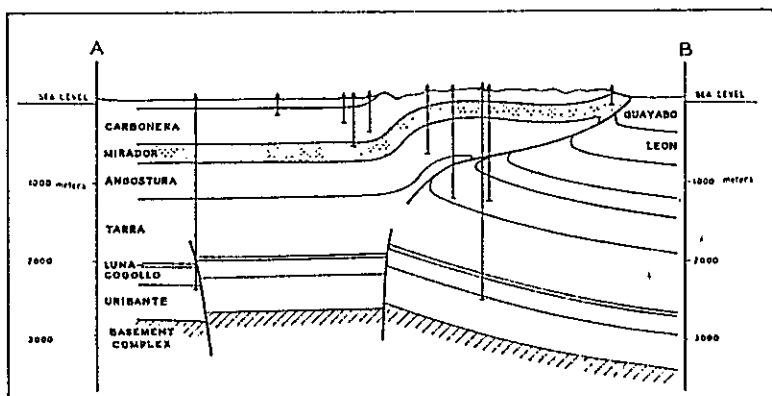


FIG. 21.—Cross section: Las Cruces field.

farther down on the north plunge of the anticline, and in 1926 the Manueles field was discovered at that place (Fig. 22). Production on a commercial scale was not possible, however, until a pipeline was constructed in 1930 to connect Cruces and Manueles with the shore of the Lake.

These two fields produce from the Angostura, Mirador, and Carbonera formations. Among these, the Mirador sandstone is highly porous and permeable, with the result that it gave an initial production of more than 4,000 barrels per day from various wells in the Manueles field. However, the same high permeability is responsible for the fact that the Mirador sandstone contains fresh water instead of oil in most parts of the Cruces field. Rain water apparently seeps into the sandstone in the mountains farther west, and then, in the eastern part of Las Cruces field, where the Mirador comes to the surface again, the water flows

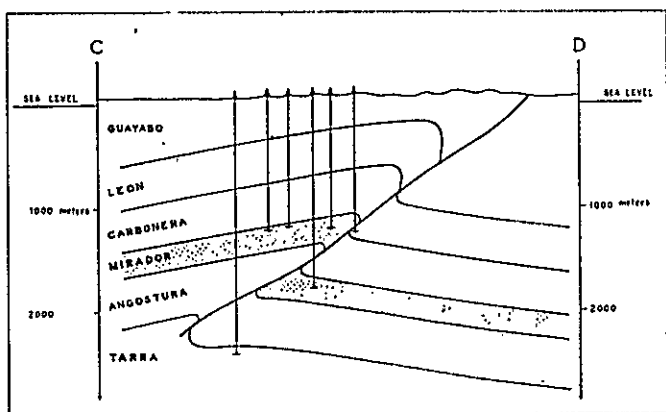


FIG. 22.—Cross section: Manueles field.

out of the formation, taking with it the oil which had been accumulated in the sandstone.

The overthrust west flank of the Cruces structures therefore produces not from the Mirador, but from thinner and less permeable sands in the Angostura and Carbonera formations where the oil seems to be trapped partly in lenses. The Mirador is productive, however, on the underthrust east flank where it is closed structurally by the thrust fault.

The Barco fields, across the border in Colombia, were known to have production from limestone in the Uribante formation of Cretaceous age. Several test wells, therefore, were drilled to that level in Las Cruces area, but no oil was found, because the structure in depth is different (Fig. 21). A structure which is closed in the Cretaceous was found eventually at West Tarra, after extensive geophysical work, and the discovery well of that field was completed in 1947 (Fig. 23).

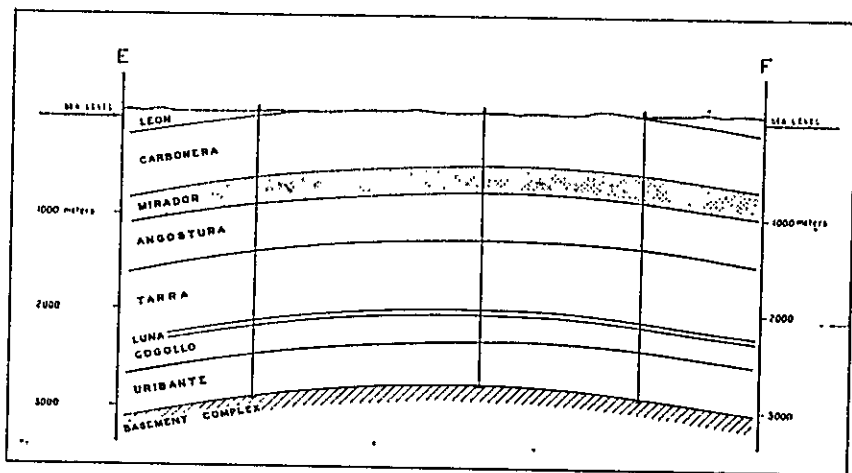


FIG. 23.—Cross section: West Tarra field.

Las Cruces, Los Manueles, and West Tarra are the only oil fields active at present on Venezuelan soil in the region southwest of Lake Maracaibo. Inactive is the Rio de Oro field, about 60 kilometers northwest of the other fields. It was discovered in 1914, by the Colon Development Company (Shell subsidiary), on a pronounced anticline with seeps. Only four wells were drilled and only very little oil was produced. No further work has been done since 1927, because of the remote location.

Further data on the Tarra fields are published in the paper by the Caribbean Petroleum Company (1948). Of interest also is the description of the Barco fields by Notestein *et al.* (1944).

4. BUCHIVACOA FIELDS

H. J. FICHTER AND H. H. RENZI¹⁴

The oil fields of the Buchivacoa District in western Falcón are relatively unimportant, with a cumulative production slightly more than 30 million barrels and a current contribution of less than 0.1 per cent of the output of all Venezuela.

The oil is of good quality, the average gravity being about 30° A.P.I. and the average gasoline content being about 30 per cent. There is a pipeline to Altamira, on the shore opposite Maracaibo.

The stratigraphic column, as known from the general area of the Buchivacoa fields, begins with the Eocene which is overlain, unconformably along folding trends, conformably in basins, by the Agua Clara (San Luis-Mitare) and Cerro Pelado formations of variable lithofacies. A minor intra-Agua Clara unconformity

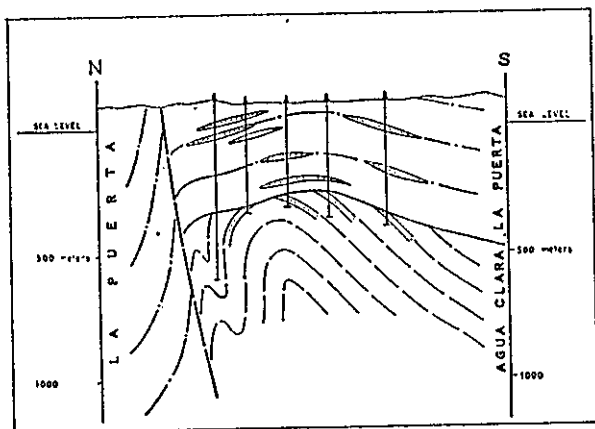


FIG. 24.—Cross section: El Mene de Mauroa field.

seemingly exists at Hombre Pintado. This sequence is cut off by the Mio-Pliocene La Puerta formation which consists of lacustrine and brackish-water sediments (mottled clays). The corresponding unconformity at the base of the La Puerta is pronounced and is the cause of the erratic occurrence of the underlying Cerro Pelado formation which is missing in some places. It indicates strong tectonic movements during Miocene time.

El Mene de Mauroa, 60 kilometers east of Maracaibo, is the oldest and most prolific of these fields. It was discovered in 1921, by the British Controlled Oilfields, Limited, relying on surface geology and the presence of seeps. There is a faulted anticline but control of the accumulation is largely stratigraphic, related to the angular unconformity between the Mio-Pliocene La Puerta formation and the Oligocene Agua Clara formation. The oil is found in sand lenses just above the unconformity, and in sandstone layers underneath, which are cut off by the unconformity (Fig. 24).

¹⁴ Shell-Caribbean Petroleum Company.

The Media field is 5 kilometers northeast of El Mene de Mauroa. It was discovered by the British Controlled Oilfields, Limited, in 1929. Oil was produced from sandstones of the Agua Clara formation, closed by faults. Only ten wells produced oil in this small field which is now abandoned.

The Hombre Pintado field is 10 kilometers farther northeast. It was discovered in 1927, jointly by the British Controlled Oilfields, Limited, and the Standard Oil Company of Venezuela (Creole). The productive formation is the Agua Clara and the accumulation is controlled by a small and complicated faulted anticline.

The Las Palmas field is 20 kilometers northeast of Hombre Pintado. It was discovered in 1928 by the Standard Oil Company of Venezuela (Creole) but is abandoned at present. The oil was obtained from sandstones of the Mitare group, of Oligocene age, on a folded and faulted structure. Only a few wells have been drilled.

More details concerning the El Mene de Mauroa, Media, and Hombre Pintado fields have been published by G. W. Halse (1947).

5. BOLÍVAR COASTAL FIELDS

W. E. WALLIS¹⁵

The Bolívar Coastal fields extend over an area straddling the eastern shore of Lake Maracaibo for a length of 69 kilometers from northwest to southeast, the width varying considerably and reaching a maximum of 16 kilometers in the central part. Active oil seeps in the area early testified to the probable existence of subsurface accumulations of petroleum and led to the drilling, by Shell, of the discovery well, completed in 1917 as a small producer from the Santa Bárbara sand of the La Rosa formation. It was, however, not until 5 years later (December 14, 1922), when the Shell well R-4 blew out, making an estimated 100,000 barrels of heavy oil per day, that the real stimulus was provided for the development of this tremendously large and rich oil-bearing area. It is a remarkable fact that, after more than 26 years of development, the limits of the productive area are not yet fully outlined.

During the earlier stages of development, drilling was carried out at widely separated localities, and the relatively long distances between production led to the belief that the area included several oil fields, to which the names Ambrosio, La Rosa, Punta Benítez, Tía Juana, Lagunillas, Pueblo Viejo, and Bachaquero were applied. Subsequent extensions of these supposedly separate "fields," however, ultimately showed that some of their reservoir sands extended from one locality to another, and that the overlap of various reservoirs was sufficient to consider the whole area as one field. When the name "Bolívar Coastal field" was applied to this whole area, it was intended that the original names should no longer be used with the word "field," but only with the word "area" to indicate

¹⁵ Creole Petroleum Corporation.

parts of the Bolívar Coastal field; however, "field" is still so commonly used in conjunction with the original names that it is now preferable to refer to the whole area as the "Bolívar Coastal fields" to avoid a paradoxical situation. Certain other names are also used to designate parts of the area, such as the PB-101 Area and LL-370 Area.

The Venezuelan Oil Concessions, a Shell subsidiary, holds all of the land concessions in the area and subsequently Shell extended its original holdings by acquiring several large Lake concessions, in one of which the exploratory well, VL-1, resulted in a new-pool discovery. The Mene Grande Oil Company, a subsidiary of the Gulf Oil Corporation, holds a number of Lake concessions embracing a one-kilometer strip adjacent to the shoreline. Creole's concessions lie entirely within the Lake.

The Bolívar Coastal fields are high on the northeastern flank of the Maracaibo basin, a large structural and topographic depression. The land overlying the productive area is very poor in outcrops, which principally consist of weathered sands and clays of Quaternary and late Tertiary age. East of Lagunillas and north of La Rosa, oil and asphalt seeps occur. The regional structure is homoclinal, the dip being gently southwest toward the center of the basin.

Within the productive area the drill has penetrated sediments ranging in age from Recent to Cretaceous. Shell's VL-1 was the first well to reach Cretaceous beds but Creole's LL-637 was the first to penetrate completely the stratigraphic section by reaching basement at 13,813 feet. There are several unconformities within the sedimentary section in the Bolívar Coastal fields; the two most pronounced occur at the top of the Paleocene and Eocene, respectively. Approximate maximum thicknesses (in feet) of the various formations making up the sections are as follows: Recent sediments, 80; El Milagro (Pleistocene), 2,100; La Puerta (Mio-Pliocene), 2,200; Lagunillas (Miocene), 2,200; La Rosa (Oligo-Miocene), 550; Icotea (Oligocene), 460; Ambrosio (Eocene), 360; Eocene above B-6-72 (Main sand), 6,200; Eocene below top of B-6-72, 10,000 est.; Guasare (Paleocene), 430; Mito Juan (Upper Cretaceous), 170; Colón (Upper Cretaceous), 1,125; Socuy (Upper Cretaceous), 65; La Luna (Upper Cretaceous), 240; Capacho (Middle Cretaceous), 480; Apon (Middle Cretaceous), 370.

The post-Eocene sediments dip gently (2° - 10°) southwest, the essentially homoclinal structure being broken by several well defined faults. The underlying Eocene erosional surface also dips southwest, but the Eocene beds themselves were folded and faulted before being subjected to truncation, these disturbances presenting a complex subsurface problem in the development of the Eocene reservoirs; such truncated Eocene highs are known to occur in the LL-370 and Pueblo Viejo areas.

At one place or another within the productive area the Lagunillas, La Rosa, Icotea, and Eocene formations have been found commercially productive. However, by far the largest volume of oil to date has come from the Lagunillas and La Rosa sands, with production from the Eocene sandstone steadily increasing dur-

ing recent years. It may be noted, however, that Eocene production did not receive serious consideration until 6-7 years ago. Cretaceous production is also a future possibility in spite of the salt-water flow from the limestone at 12,106 feet in Shell's VL-1 and the non-productive section found in LL-637.

Accumulations of oil and gas in the post-Eocene sediments are primarily confined to stratigraphic traps formed by the updip pinching-out of the Miocene

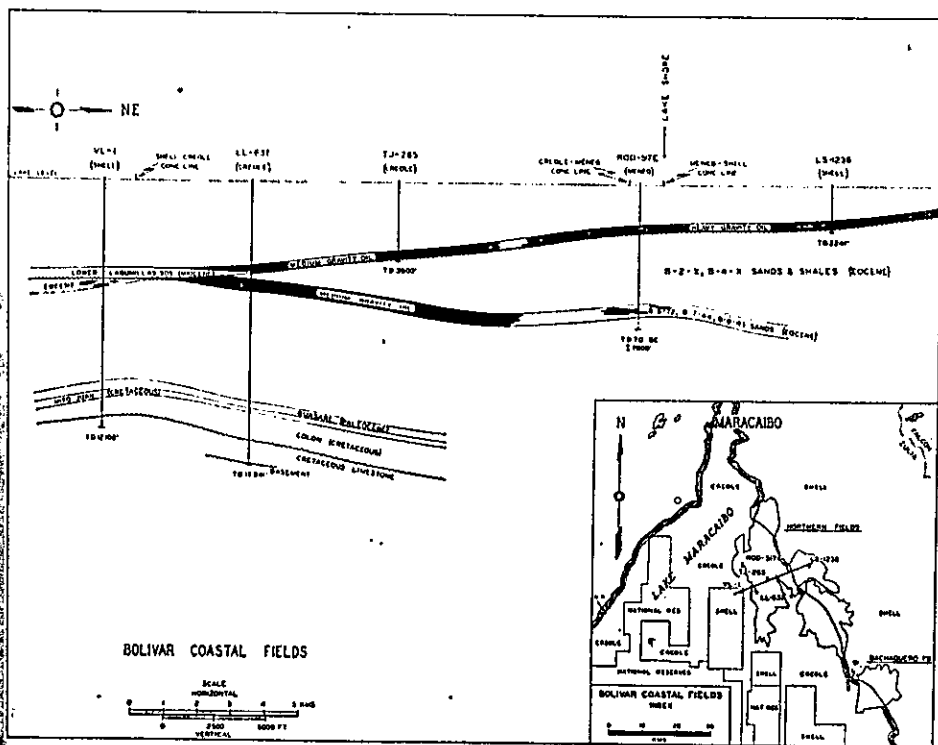


Fig. 25.—Diagrammatic cross section: Bolivar Coastal fields.

sands or by sealing of these sands at their outcrops by tar oil, local structure playing only a minor role (Figs. 25 and 26). The Ico tea sand (Oligocene) is found only in local lows of the Eocene erosion surface and thus also contains only stratigraphic accumulations. In the underlying Eocene beds, accumulations are largely controlled by truncation and overlap.

The oil produced varies in gravity from 11° to 34° API. The crude is of mixed and asphaltic base with a wide range of characteristics and is segregated into several streams.

Accumulative total production from the Bolivar Coastal fields, to July 1, 1950, was 3,282,951,000 barrels.

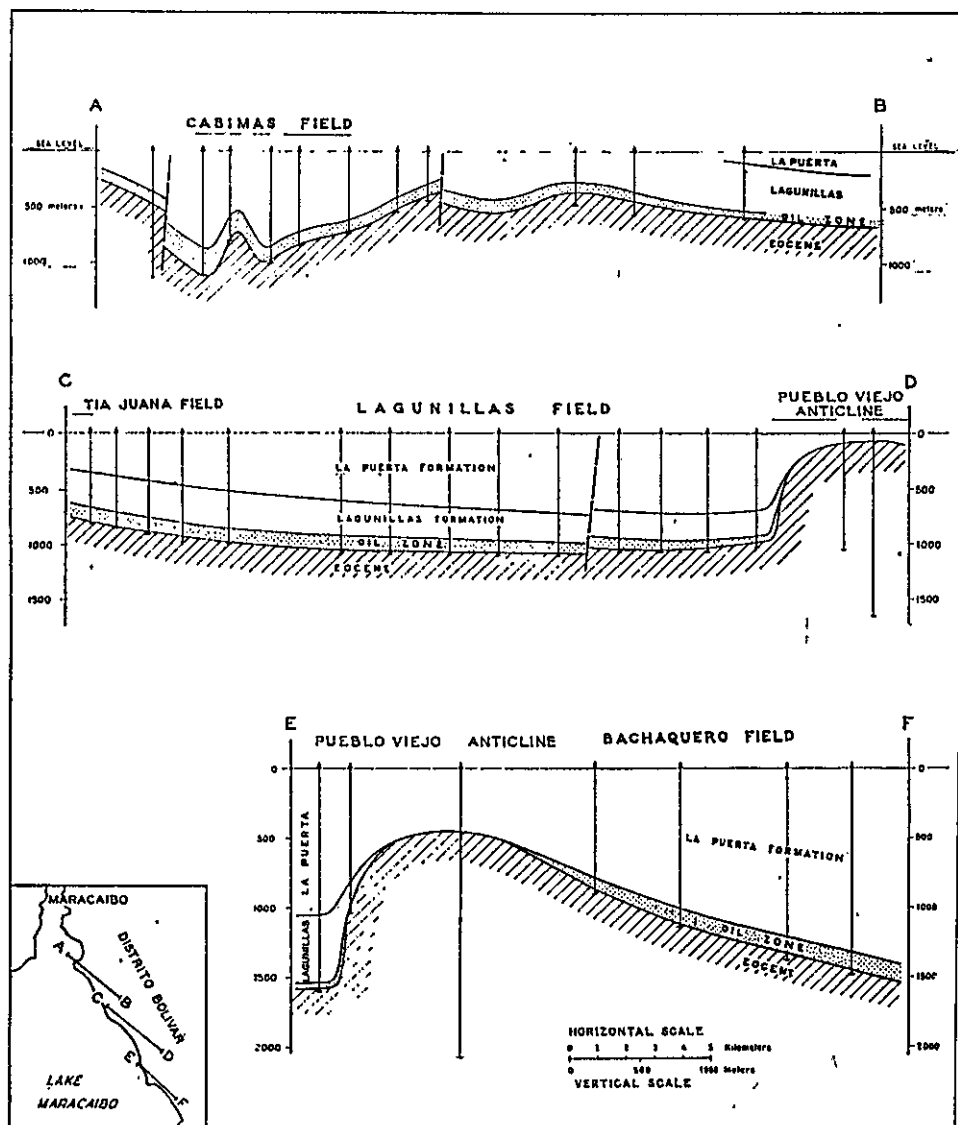


FIG. 26.—Cross sections: Bolívar Coastal fields.

For a more detailed description of the Bolívar Coastal fields, the reader is referred to a paper by the Caribbean Petroleum Company (1948) and to that of Sutton (1946).

6. CUMAREBO FIELD

W. E. WALLIS¹⁸

The Cumarebo field, discovered in 1931, is an elongate anticlinal dome striking northeast, and located on the northwest-dipping flank of the Ricoa uplift (Fig. 27). The Cumarebo structure was first delineated by surface geology, and the wildcat well which discovered the field, CU-1, a proposed 5,000-foot test drilled on the crest of the anticline, encountered the top of Sand 8 (Caujarao formation—middle Miocene) at 610 feet, and was completed with an initial potential of 700 barrels per day at total depth of 627 feet.

Folding of the structure probably took place in late Miocene-early Pliocene time, resulting in a northeast-plunging anticline with a long axis of 4 kilometers and an average width of oil accumulation of 850 meters. The dome is severely faulted, there being three systems of faults (regional, secondary, and strike) represented. Of these, the large regional, northwest-striking faults, which dip approximately 45° northeast and apparently occurred prior to the folding of the anticline, are of major importance in that they form barriers and blocks which have controlled accumulation of petroleum in the area. The structure is bordered on the southeast by the Taica syncline—a complementary feature whose axis lies 1,700 meters southeast of the axis of the anticline. The long, rather steeply dipping northwest flank extends into the regional deep on the northwest, from which direction the oil accumulation, derived from the dark gray, ordinarily pure shales of the Caujarao and Mosquito formations, probably came.

Rocks of Miocene age form the outcropping and producing formations of the field, and Mio-Oligocene formations have been penetrated in deep tests. Oil is produced from the fine-grained, fairly well sorted sands of the Caujarao and Mosquito formations, there being a total of 13 producing sands in the field. However, most of the sands are relatively thin and locally productive, the bulk of the oil being produced from Sands 10 and 12 of the Caujarao, and Sand 15 of the Mosquito formation. Depth of the producing beds varies from 405 feet (Sand 6 in CU-67) to 2,635 feet (Sand 16 in CU-21).

As of June 30, 1950, Cumarebo had produced 41,725,000 barrels of 48° API gravity oil from 550 proved surface acres. In all, 133 wells were drilled. Present production averages 3,400 barrels per day from 42 wells.

A detailed description of the Cumarebo field has recently been published by Payne (1951).

¹⁸ Creole Petroleum Corporation.